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Euclid's Book on Divisions of Figures

Euclid, Raymond Clare Archibald, Franz Woepcke, and Leonardo Fibonacci

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EUCLID’S BOOK ON DIVISIONS OF FIGURES

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EUCLID’S BOOK ON DIVISIONS OF FIGURES (περὶ διαιρέσεων βιβλίον) WITH A RESTORATION BASED ON WOEPCKE’S TEXT AND ON THE PRACTICA GEOMETRIAE OF LEONARDO PISANO BY RAYMOND CLARE ARCHIBALD, Ph.D. ASSISTANT PROFESSOR OF MATHEMATICS IN BROWN UNIVERSITY, PROVIDENCE, RHODE ISLAND Cambridge: at the University Press 1915.

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TO MY OLD TEACHER AND FRIEND ALFRED DEANE SMITH PROFESSOR OF GREEK AND LATIN AT MOUNT ALLISON UNIVERSITY FOR FORTY-FOUR YEARS SCHOLAR OF GREAT ATTAINMENTS THE WONDER OF ALL WHO KNOW HIM THESE PAGES ARE AFFECTIONATELY DEDICATED

INTRODUCTORY Euclid, famed founder of the Alexandrian School of Mathematics, was the author of not less than nine works. Approximately complete texts, all carefully edited, of four of these, (1) theElements, (2) theData, (3) the Optics, (4) thePhenomena, are now our possession. In the case of (5) the Pseudaria, (6) the Surface-Loci, (7) the Conics, our fragmentary knowl- edge, derived wholly from Greek sources, makes conjecture as to their content of the vaguest nature. On (8) thePorisms, Pappus gives extended comment. As to (9), the book On Divisions (of figures), Proclus alone among Greeks makes explanatory reference. But in an Arabian MS., trans- lated by Woepcke into French over sixty years ago, we have not only the

enunciations of all of the propositions but also the proofs of four of them. Whilst elaborate restorations of thePorisms by Simson and Chasles have been published, no previous attempt has been made (the pamphlet of Ofterdinger is not forgotten) to restore the proofs of the bookOn Divisions (of figures). And, except for a short sketch in Heath's monumental edition of Euclid's Elements, nothing but passing mention of Euclid's bookOn Divisions has appeared in English. In this little volume I have attempted: (1) to give, with necessary commentary, a restoration of Euclid's work based on the Woepcke text and on a thirteenth century geometry of Leonardo Pisano. (2) to take due account of the various questions which arise in connection with (a) certain MSS. of "Muhammed Bagdedinus," (b) the Dee- Commandinus book on divisions of figures. (3) to indicate the writers prior to 1500 who have dealt with propositions of Euclid's work. (4) to make a selection from the very extensive bibliography of the sub- ject during the past 400 years. In the historical survey the MSS. of "Muhammed Bagdedinus" play an important rôle, and many recent historians, for example Heiberg, Cantor, Hankel, Loria, Suter, and Steinschneider, have contributed to the discus- sion. As it is necessary for me to correct errors, major and minor, of all of these writers, considerable detail has to be given in the first part of the volume; the brief second part treats of writers on divisions before 1500; the third part contains the restoration proper, with its thirty-six propositions. The Appendix deals with literature since 1500. A score of the propositions are more or less familiar as isolated problems of modern English texts, and are also to be found in many recent English,

INTRODUCTORY vii German and French books and periodicals. But any approximately accu- rate restoration of the work as a whole, in Euclidean manner, can hardly fail of appeal to anyone interested in elementary geometry or in Greek mathematics of twenty-two centuries ago. In the spelling of Arabian names, I have followed Suter. It is a pleasure to have to acknowledge indebtedness to the two foremost living authorities on Greek Mathematics. I refer to Professor J. L. Heiberg of the University of Copenhagen and to Sir Thomas L. Heath of London. Professor Heiberg most kindly sent me the proof pages of the forthcoming concludingvolumeofEuclid's Opera Omnia, whichcontainedthereferences to Euclid's bookOn Divisions of Figures. To Sir Thomas my debt is great. On nearly every page that follows there is evidence of the influence of his publications; moreover, he has read this little book in proof and set me right at several points, more especially in connection with discussions in Note 113 and Paragraph 50. R. C. A. Brown University, June, 1915.

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Apollonius’s treatiseOn Cutting off a Spaceand a Pappus-lemma to Euclid’s book ofPorisms 24 III 22–57 Restoration of Euclid’sπερὶ διαιρέσεων βιβλίον 27 IV Appendix 79 Index of Names 91

I. Proclus, and Euclid’s book On Divisions. 1. Last in a list of Euclid’s works “full of admirable diligence and skilful consideration,” Proclus mentions, without comment,περὶ διαιρέσεων βιβλίον 1. But a little later2 in speaking of the conception or definition offigure and of the divisibility of a figure into others differing from it in kind, Proclus adds: “For the circle is divisible into parts unlike in definition or notion, and so is each of the rectilineal figures; this is in fact the business of the writer of the Elements in his Divisions, where he divides given figures, in one case into like figures, and in another into unlike3.” De Divisionibus by Muhammed Bagdedinus and the Dee MS. 2. This is all we have from Greek sources, but the discovery of an Arabian translation of the treatise supplies the deficiency. In histories of Euclid’s works (for example those by Hankel4, Heiberg5, Favaro6, Loria7, Cantor8, Hultsch9, Heath3) prominence is given to a treatiseDe Divisionibus, by one “Muhammed Bagdedinus.” Of this in 156310 a copy (in Latin) was given by John Dee to 1 Procli Diadochi in primum Euclidis elementorum librum commentarii ex rec. G. Friedlein, Leipzig, 1873, p. 69. Reference to this work will be made by “Proclus.” 2 Proclus1, p. 144. 3 In this translation I have followedT. L. Heath, The Thirteen Books of Euclid’s Elements, 1, Cambridge, 1908, p. 8. To Heath’s account (pp. 8–10) of Euclid’s bookOn Divisions I shall refer by “Heath.” “Like” and “unlike” in the above quotation mean, not “similar” and “dissimilar” in the technical sense, but “like” or “unlike in definition or notion”: thus to divide a triangle into triangles would be to divide it into “like” figures, to divide a triangle into a triangle and a quadrilateral would be to divide it into “unlike” figures. (Heath.) 4 H. Hankel, Zur Geschichte der Mathematik, Leipzig, 1874, p. 234. 5 J. L. Heiberg, Litterargeschichtliche Studien über Euklid, Leipzig, 1882, pp. 13– 16, 36–38. Reference to this work will be made by “Heiberg.” 6 E. A. Favaro. “PreliminariadunaRestituzionedellibrodiEuclidesulladivisione delle figure piane,” Atti del reale Istituto Veneto di Scienze, Lettere ed Arti, i6, 1883, pp. 393–6. “Notizie storico-critiche sulla Divisione delee Aree” (Presentata li 28 gennaio, 1883), Memorie del reale Istituto Veneto di Scienze, Lettere ed Arti,xxii, 129–154. This is by far the most elaborate consideration of the subject up to the present. Reference to it will be made by “Favaro.” 7 G. Loria, “Le Scienze esatte nell’ antica Grecia, Libroii, Il periodo aureo della geometriaGreca.” Memorie della regia Accademia di Scienze, Lettere ed Arti in Modena, xi2, 1895, pp. 68–70, 220–221.Le Scienze esatte nell’ antica Grecia, Seconda edizione. Milano, 1914, pp. 250–252, 426–427. 8 M. Cantor, Vorlesungen über Geschichte der Mathematik, i3, 1907, pp. 287–8; ii2, 1900, p. 555. 9 F. Hultsch, Article “Eukleides” in Pauly-Wissowa’sReal-Encyclopädie der Class. Altertumswissenschaften, vi, Stuttgart, 1909, especially Cols. 1040–41. 10 When Dee was in Italy visiting Commandinus at Urbino.

2 EUCLID’S BOOK ON DIVISION OF FIGURES I [2 Commandinus who published it in Dee’s name and his own in 157011. Recent writers whose publications appeared before 1905 have generally supposed that Dee had somewhere discovered an Arabian original of Muhammed’s work and had given a Latin translation to Commandinus. Nothing contrary to this is in- deed explicitly stated by Steinschneider when he writes

in 190512, “Machomet Bagdadinus (=aus Bagdad) heisst in einem alten MS. Cotton (jetzt im Brit. Mus.) der Verfasser von: de Superficierum divisione (22 Lehrsätze); Jo. Dee aus London entdeckte es und übergab es T. Commandino....” For this suggestion as to the place where Dee found the MS. Steinschneider gives no authority. He does, however, give a reference to Wenrich13, who in turn refers to a list of the printed books (“Impressi”) of John Dee, in a life of Dee by Thomas Smith14 (1638–1710). We here find as the third in the list, “Epistola ad eximium Ducis Urbini Mathematicum, Fredericum Commandinum, praefixa libello Machometi 11 De superficierum divisionibus liber Machometo Bagdedino ascriptus nunc pri- mum Joannis Dee Londinensis & Federici Commandini Urbinatis opera in lucem ed- itus. Federici Commandini de eadem re libellus. Pisauri, mdlxx. In the same year appeared an Italian translation: Libro del modo di dividere le superficie attribuito a Machometo Bagdedino. Mandato in luce la prima volta da M. G. Dee...e da M. F....Commandino...Tradotti dal Latino in volgare da F. Viani de’ Malatesti,... In Pesaro, delmdlxx...4 unnumbered leaves and 44 numbered on one side. An English translation from the Latin, with the following title-page, was published in the next century:A Book of the Divisions of Superficies: ascribed to Machomet Bagde- dine. Now put forth, by the pains of John Dee of London, and Frederic Commandine of Urbin. As also a little Book of Frederic Commandine, concerning the same matter. London Printed by R. & W. Leybourn, 1660. Although this work has a separate title page and the above date, it occupies the last fifty pages (601–650) of a work dated a year later:Euclid’s Elements of Geometry in XV Books...to which is added a Treatise of Regular Solids by Campane and Flussas likewise Euclid’s Data and Marinus Preface thereunto annexed. Also a Treatise of the Divisions of Superficies ascribed to Machomet Bagdedine, but published by Commandine, at the request of John Dee of London; whose Preface to the said Treatise declares it to be the Worke of Euclide, the Author of the Elements. Published by the care and Industry of John Leeke and George Serle, Students in the Mathematics. London... mdclxi. A reprint of simply that portion of the Latin edition which is the text of Muhammed’s work appeared in:ΕΥΚΛΕΙΔΟΥ ΤΑ ΣΩΖΟΜΕΝΑ Euclidis quae supersunt omnia. Ex rescensione Davidis Gregorii...Oxoniae... mdcciii. Pp. 665–684: ΕΥΚΛΕΙΔΟΥ ΩΣ ΟΙΟΝΤΑΙ ΤΙΝΕΣ, ΠΕΡΙ ΔΙΑΙΡΕΣΕΩΝ ΒΙΒΛΟΣ Euclidis, ut quidam arbitrantur, de divisionibus liber—vel ut alii volunt, Machometi Bagdedini liber de divisionibus superficierum.” 12 M. Steinschneider, “Die Europäischen Übersetzungen aus dem Arabischen bis Mitte des 17. Jahrhunderts.” Sitzungsberichte der Akademie der Wissenschaften in Wien (Philog.-histor. Klasse) cli, Jan. 1905, Wien, 1906. Concerning “171. Muham- med” cf. pp. 41–2. Reference to this paper will be made by “Steinschneider.” 13 J. G. Wenrich, De auctorum Graecorum versionibus. Lipsiae, mdcccxlii, p. 184. 14 T. Smith, Vitae quorundam eruditissimorum et illustrium virorum... Lon- dini... mdccvii, p. 56. It was only the first 55 pages of this “Vita Joannis Dee, Mathe- matici Angli,” which were translated into English by W. A. Ayton, London, 1908.

3–4] MSS. OF MUHAMMED BAGDEDINUS AND DEE 3 Bagdedini de superficierum divisionibus...Pisauri, 1570. Exstat MS. in Biblio- theca Cottoniana sub Tiberio Bix.” Then come the following somewhat mysterious sentences which I give in translation15: “After the preface Lord Ussher [1581–1656], Archbishop of Ar- magh,

has these lines: It is to be noted that the author uses Euclid’s Ele- ments translated into the Arabic tongue, which Campanus afterwards turned into Latin. Euclid therefore seems to have been the author of the Propositions [of De Divisionibus] though not of the demonstrations, which contain references to an Arabic edition of the Elements, and which are due to Machometus of Bagded or Babylon.” This quotation from Smith is reproduced, with various changes in punctuation and typography, by Kästner16. Consideration of the latter part of it I shall postpone to a later article (5). 3. FollowingupthesuggestionofSteinschneider, Suterpointedout 17, with- out reference to Smith14 or Kästner16, that in Smith’s catalogue of the Cotto- nian Library there was an entry18 under “Tiberius19 Bix, 6”: “Liber Divisionum Mahumeti Bag-dadini.” As this MS. was undoubtedly in Latin and as Cottonian MSS. are now in the British Museum, Suter inferred that Dee simply made a copy of the above mentioned MS. and that this MS. was now in the British Museum. With his wonted carefulness of statement, Heath does not commit himself to these views although he admits their probable accuracy. 4. As a final settlement of the question, I propose to show that Steinschnei- der and Suter, and hence also many earlier writers, have not considered all facts available. Some of their conclusions are therefore untenable. In particular: (1) In or before 1563 Dee didnot make a copy of any Cottonian MS.; (2) The above mentioned MS. (Tiberius, B.ix, 6) was never, in its entirety, in the British Museum; 15 “Post praefationem haec habet D.Usserius Archiepiscopus Armachanus. Notan- dum est autem, Auctorem hunc Euclide usum in Arabicam linguam converso, quem postea Campanus Latinum fecit. Auctor igitur propositionum videtur fuisseEuclides: demonstrationum, in quibus Euclides in Arabico codice citatur, Machometus Bagded sive Babylonius.” It has been stated that Campanus (13. cent.) did not translate Euclid’s Elements into Latin, but that the work published as his (Venice, 1482—the first printed edition of theElements) was the translation made about 1120 by the English monk Athelhard of Bath. Cf. Heath, Thirteen Books of Euclid’s Elements, i, 78, 93–96. 16 A. G. Kästner, Geschichte der Mathematik... Erster Band...Göttingen, 1796, pp. 272–3. See also “Zweyter” Band, 1797, pp. 46–47. 17 H. Suter, “Zu dem Buche ‘De Superficierum divisionibus’ des Muhammed Bagdedinus.” Bibliotheca Mathematica, vi3, 321–2, 1905. 18 T. Smith, Catalogus Librorum Manuscriptorum Bibliothecae Cottonianae...Ox- onii,... mdcxcvi, p. 24. 19 The original Cottonian library was contained in 14 presses, above each of which was a bust; 12 of these busts were of Roman Emperors. Hence the classification of the MSS. in the catalogue.

4 EUCLID’S BOOK ON DIVISION OF FIGURES I [4 (3) The inference by Suter that this MS. was probably the Latin translation of the tract from the Arabic, made by Gherard of Cremona (1114–1187)—among the lists of whose numerous translations a “liber divisionum” occurs—should be accepted with great reserve; (4) The MS. which Dee used can be stated with absolute certainty and this MS. did not, in all probability, afterwards become a Cottonian MS. (1) Sir Robert Bruce Cotton, the founder of the Cottonian Library, was born in 1571. The Cottonian Library was not, therefore, in existence in 1563 and Dee could not then have copied a Cottonian MS. (2) The Cottonian Library passed into the care of the nation shortly after 1700. In 1731 about 200 of the MSS. were damaged or destroyed by fire. As a result of the parliamentary

inquiry Casley reported20 on the MSS. destroyed or injured. Concerning Tiberiusix, he wrote, “This volume burnt to a crust.” He gives the title of each tract and the folios occupied by each in the volume. “Liber Divisionum Mahumeti Bag-dadini” occupied folios 254–258. When the British Museum was opened in1753, what was left of the Cottonian Library was immediately placed there. Although portions of all of the leaves of our tract are now to be seen in the British Museum, practically none of the writing is decipherable. (3) Planta’s catalogue21 has the following note concerning Tiberiusix: “A volumeonparchment, whichonceconsistedof272leaves, writtenabouttheXIV. century [not the XII. century, when Gherard of Cremona flourished], containing eight tracts, the principal of which was a ‘Register of William Cratfield, abbot 20 D. Casley, p. 15ff. ofA Report from the Committee appointed to view the Cot- tonian Library...Published by order of the House of Commons. London, mdccxxxii (British Museum MSS. 24932). Cf. also the page opposite that numbered 120 inA Catalogue of the Manuscripts in the Cottonian Library...with an Appendix contain- ing an account of the damage sustained by the Fire in 1731; by S. Hooper... Lon- don:... mdcclxxvii. 21 J. Planta, A Catalogue of the Manuscripts in the Cottonian Library deposited in the British Museum. Printed by command of his Majesty King George III...1802. In the British Museum there are three MS. catalogues of the Cottonian Library: (1) Harleian MS. 6018, a catalogue made in 1621. At the end are memoranda of loaned books. On a sheet of paper bearing date Novem. 23, 1638, Tiberius Bix is listed (folio 187) with its art. 4: “liber divisione Machumeti Bagdedini.” The paper is torn so that the name of the person to whom the work was loaned is missing. The volume is not mentioned in the main catalogue. (2) MS. No.36789, made after Sir Robert Cotton’s death in 1631 and before 1638 (cf. Catalogue of Additions to the MSS. in British Museum, 1900–1905...London, 1907, pp. 226–227), contains, apparently, no reference to “Muhammed.” (3) MS. No. 36682 A, of uncertain date but earlier than 1654 (Catalogue of Addi- tions...l.c. pp. 188–189). On folio 78 verso we find Tiberius B ix, Art. 4: “Liber divisione Machumeti Bagdedini.” A “Muhammed” MS. was therefore in the Cottonian Library in 1638. The anonymously printed (1840?) “Index to articles printed from the Cotton MSS., & where they may be found” which may be seen in the British Museum, only gives references to the MSS. in “Julius.”

5] MSS. OF MUHAMMED BAGDEDINUS AND DEE 5 of St Edmund’” [d. 1415]. Tracts 3, 4, 5 were on music. (4) On “A◦ 1583, 6 Sept.” Dee made a catalogue of the MSS. which he owned. This catalogue, which is in the Library of Trinity College, Cambridge22, has been published23 under the editorship of J. O. Halliwell. The 95th item described is a folio parchment volume containing 24 tracts on mathematics and astronomy. The 17th tract is entitled “Machumeti Bagdedini liber divisionum.” As the contents of this volume are entirely different from those of Tiberiusix described above, in (3), it seems probable that there were two copies of “Muhammed’s” tract, while the MS. which Dee used for the 1570 publication was undoubtedly his own, as we shall presently see. If the two copies be granted, there is no evidence against the Dee copy having been that made by Gherard of Cremona. 5. There is the not remote possibility that the Dee MS. was destroyed soon after it was catalogued. For in the same month that the above catalogue was prepared, Dee left his home at Mortlake, Surrey, for a lengthy trip in Europe. Immediately after his departure “the mob, who

execrated him as a magician, broke into his house and destroyed a great part of his furniture and books24...” many of which “were the written bookes25.” Now the Dee catalogue of his MSS. (MS. O. iv. 20), in Trinity College Library, has numerous annotations26 in Dee’s handwriting. They indicate just what works were (1) destroyed or stolen (“Fr.”)27 and (2) left(“T.”)28 after the raid. Opposite the titles of the tracts in the volume including the tract “liber divisionum,” “Fr.” is written, and opposite the title “Machumeti Bagdedini liber divisionum” is the following note: “Curavi imprimi Urbini in Italia per Federicum Commandinum exemplari descripto ex vetusto isto monumento(?) per me ipsum.” Hence, as stated above, it is now definitely known (1) that the MS. which Dee used was his own, and (2) that some 20 years after he made a copy, the MS. was stolen and probably destroyed29. On the other hand we have the apparently contradictory evidence in the passage quoted above (Art. 2) from the life of Dee by Smith14 who was also the compiler of the Catalogue of the Cottonian Library. Smith was librarian when he wrote both of these works, so that any definite statement which he 22 A transcription of the Trinity College copy, by Ashmole, is in MS. Ashm. 1142. Another autograph copy is in the British Museum: Harleian MS. 1879. 23 Camden Society Publications, xix, London,m.dccc.xlii. 24 Dictionary of National Biography, Article, “Dee, John.” 25 “The compendious rehearsall of John Dee his dutifull declaration A. 1592” printed in Chetham Miscellanies, vol.i, Manchester,1851, p. 27. 26 Although Halliwell professed to publish the Trinity MS., he makes not the slightest reference to these annotations. 27 “Fr.” is no doubt an abbreviation forFuratum. 28 “T.”, according to Ainsworth (Latin Dictionary), was put after the name of a soldier to indicate that he had survived (superstes). Whence this abbreviation? 29 The view concerning the theft or destruction of the MS. is borne out by the fact that in a catalogue of Dee’s Library (British Museum MS. 35213) made early in the seventeenth century (Catalogue of Additions and Manuscripts...1901, p. 211), Machumeti Bagdedini is not mentioned.

 [6 makes concerning the library long in his charge is not likely to be successfully challenged. Smith does not however say that Dee’s “Muhammed” MS. was in the Cottonian Library, and if he knew that such was the case we should certainly expect some note to that effect in the catalogue18; for in three other places in his catalogue (Vespasian Bx, Aii13, Galba Eviii), Dee’s original ownership of MSS. which finally came to the Cottonian Library is carefully remarked. Smith does declare, however, that the Cottonian MS. bore, “after the preface,” certain notes (which I have quoted above) by Archbishop Ussher (1581–1656). Now it is not a little curious that these notes by Ussher, who was not born till after the Dee book was printed, should be practically identical with notes in the printed work, just after Dee’s letter to Commandinus (Art. 3). For the sake of comparison I quote the notes in question30; “To the Reader.—I am here to advertise thee (kinde Reader) that this author which we present to thee, made use of Euclid translated into the Arabick Tongue, whom afterwards Campanus made to speake Latine. This I thought fit to tell thee, that so in searching or examining the Propositions which are cited by him, thou mightest not sometime or other trouble thy selfe in vain, Farewell.” The Dee MS. as published did not have any preface. We can therefore only assume that Ussher wrote in a MS. whichdid have a preface the few

lines which he may have seen in Dee's printed book. 6. Other suggestions which have been made concerning "Muhammed's" tract should be considered. Steinschneider asks, "Ob identisch de Curvis super- ficiebus, von einem Muhammed, MS. Brit. Mus. Harl. 6236 (i, 191)31?" I have examined this MS. and found that it has nothing to do with the subject matter of the Dee tract. But again, Favaro states32: "Probabilmente il manoscritto del quale si servì il Dee è lo stesso indicato dall'Heilbronner33 come esistente nella Biblioteca Bodleiana di Oxford." Under date "6. 3. 1912" Dr A. Cowley, assistant librarian in the Bodleian, wrote me as follows: "We do not possess a copy of Heilbronner's Hist. Math. Univ. In the old catalogue of MSS. which he would have used, the work you mention is included—but is really a printed book and is only included in the catalogue of MSS. because it contains some manuscript notes— "Its shelf-mark is Savile T 20. "It has 76 pages in excellent condition. The title page has: De Superficierum | divisionibus liber| Machometo Bagdedino| ascriptus| nunc primum Joannis Dee| ...| opera in lucem editus| ...Pisauri mdlxx. 30 This quotation from the Leeke-Serle Euclid11 is an exact translation of the original. 31 This should be 6256 (i, 391). 32 Favaro, p. 140. Cf. Heiberg, p. 14. This suggestion doubtless originated with Ofterdinger38, p. [1]. 33 J. C. Heilbronner, Historia matheseos Universae...Lipsiae, mdccxlii, p. 620: ("Manuscripta mathematica in Bibliotheca Bodlejana") "34 Mohammedis Bagdadeni liber de superficierum divisionibus, cum Notis H. S."

7] THE WOEPCKE-EUCLID MS. 7 "The MS. notes are by Savile, from whom we got the collection to which this volume belongs." The notes were incorporated into the Gregory edition11 of the Dee tract. Here and elsewhere34 Savile objected to attributing the tract to Euclid as au- thor35. His arguments are summed up, for the most part, in the conclusions of Heiberg followed by Heath: "the Arabic original could not have been a direct translation from Euclid, and probably was not even a direct adaptation of it; it contains mistakes and unmathematical expressions, and moreover does not contain the propositions about the division of a circle alluded to by Proclus. Hence it can scarcely have contained more than a fragment of Euclid's work." The Woepcke-Euclid MS. 7. On the other hand Woepcke found in a MS. (No. 952. 2 Arab. Suppl.) of the Bibliothèque nationale, Paris, a treatise in Arabic on the division of plane figures, which he translated, and published in 185136. "It is expressly 34 H. Savile, Praelectiones tresdecim in principium elementorum Evclidis, Oxonii habitae M.DC.XX. Oxonii..., 1621, pp. 17–18. 35 Dee's statement of the case in his letter to Commandinus (Leeke-Serle Euclid11, cf. note 30) is as follows: "As for the authors name, I would have you understand, that to the very old Copy from whence I writ it, the name ofMachomet Bagdedinewas put in ziphers or Characters, (as they call them) who whether he were thatAlbategnus whom Copernicus often cites as a very considerable Author in Astronomie; or that Machomet who is said to have beenAlkindus's scholar, and is reported to have written somewhat of the art of Demonstration, I am not yet certain of: or rather that this may be deemed a Book of ourEuclide, all whose Books were long since turned out of the Greeke into the Syriack and Arabick Tongues. Whereupon, It being found some time or other to want its Title with theArabians or Syrians, was easily attributed by the transcribers to that most famous Mathematician among them, Machomet: which I am able to prove by many testimonies, to be often done in many Moniments of the Ancients; ...yea further, we

could not yet perceive so great acuteness of anyMachomet in the Mathematicks, from their moniments which we enjoy, as everywhere appears in these Problems. Moreover, thatEuclide also himself wrote one Bookπερι διαιρέσεων that is to say,of Divisions, as may be evidenced from Proclus's Commentaries upon his first of Elements: and we know none other extant under this title, nor can we find any, which for excellencie of its treatment, may more rightfully or worthily be ascribed toEuclid. Finally, I remember that in a certain very ancient piece of Geometry, I have read a place cited out of this little Book in expresse words, even as from amost (sic) certain work of Euclid. Therefore we have thus briefly declared our opinions for the present, which we desire may carry with them so much weight, as they have truth in them.... But whatsoever that Book ofEuclid was concerning Divisions, certainly this is such an one as may be both very profitable for the studies of many, and also bring much honour and renown to every most noble ancient Mathematician; for the most excellent acutenesse of the invention, and the most accurate discussing of all the Cases in each Probleme...." 36 F. Woepcke, "Notice sur des traductions Arabes de deux ouverages perdus d'Euclide" Journal Asiatique, Septembre–Octobre, 1851,xviii4, 217–247. Euclid's work On the division (of plane figures): pp. 233–244. Reference to this paper will be made by "Woepcke." InEuclidis opera omnia, vol. 8, now in the press, there are "Fragmenta

8 EUCLID'S BOOK ON DIVISION OF FIGURES I [8–9 attributed to Euclid in the MS. and corresponds to the description of it by Proclus. Generally speaking, the divisions are divisions into figures of the same kind as the original figures, e. g. of triangles into triangles; but there are also divisions into 'unlike' figures, e. g. that of a triangle by a straight line parallel to the base. The missing propositions about the division of a circle are also here: 'to divide into two equal parts a given figure bounded by an arc of a circle and two straight lines including a given angle' and 'to draw in a given circle two parallel straight lines cutting off a certain part of a circle.' Unfortunately the proofs are given of only four propositions (including the two last mentioned) out of 36, because the Arabian translator found them too easy and omitted them." That the omission is due to the translator and did not occur in the original is indicated in two ways, as Heiberg points out. Five auxiliary propositions (Woepcke 21, 22, 23, 24, 25) of which no use is made are introduced. Also Woepcke 5 is: "...and we divide the triangle by a construction analogous to the preceding construction"; but no such construction is given. The four proofs that are given are elegant and depend only on the proposi- tions (or easy deductions from them) of theElements, while Woepcke 18 has the true Greek ring: "to apply to a straight line a rectangle equal to the rectangle contained byAB, AC and deficient by a square." 8. To no proposition in the Dee MS. is there word for word correspon- dence with the propositions of Woepcke but in content there are several cases of likeness. Thus, Heiberg continues, Dee 3 = Woepcke 30 (a special case is Woepcke 1); Dee 7 = Woepcke 34 (a special case is Woepcke 14); Dee 9 = Woepcke 36 (a special case is Woepcke 16); Dee 12 = Woepcke 32 (a special case is Woepcke 4). Woepcke 3 is only a special case of Dee 2; Woepcke 6, 7, 8, 9 are easily solved by Dee 8. And it can hardly be chance that the proofs of exactly these propositions in Dee should be without fault. That the treatise published by Woepcke is no fragment but the

complete work which was before the translator is expressly stated37, “fin du traité.” It is moreover a well ordered and compact whole. Hence we may safely conclude that Woepcke’s is not only Euclid’s own work but the whole of it, except for proofs of some propositions. collegit et disposuit J. L. Heiberg,” through whose great courtesy I have been enabled to see the proof-sheets. First among the fragments, on pages 227–235, are (1) the Proclus references to περι διαιρέσεων and (2) the Woepcke translation mentioned above. In the article on Euclid in the last edition of theEncyclopaedia Britannicano reference is made to this work or to the writings of Heiberg, Hultsch, Steinschneider and Suter. 37 Woepcke, p. 244.

10–11] PRACTICA GEOMETRIAE OF LEONARDO PISANO 9 9. For the reason just stated the so-calledWiederherstellung of Euclid’s work by Ofterdinger38, based mainly on Dee, is decidedly misnamed. A more accurate description of this pamphlet would be, “A translation of the Dee tract withindicationsinnotesofacertaincorrespondencewith15ofWoepcke’spropo- sitions, the whole concluding with a translation of the enunciations of 16 of the remaining 21 propositions of Woepcke not previously mentioned.” Woepcke 30, 31, 34, 35, 36 are not even noticed by Ofterdinger. Hence the claim I made above (“Introductory”) that the first real restoration of Euclid’s work is now presented. Having introduced Woepcke’s text as one part of the basis of this restoration, the other part demands the consideration of the Practica Geometriae of Leonardo Pisano (Fibonaci). 10. It was in the year 1220 that Leonardo Pisano, who occupies such an important place in the history of mathematics of the thirteenth century39, wrote his Practica Geometriae, and the MS. is now in the Vatican Library. Although it was known and used by other writers, nearly six and one half centuries elapsed before it was finally published by Prince Boncompagni40. Favaro was the first6 to call attention to the importance of Section IIII41 of the Practica Geome- triae in connection with the history of Euclid’s work. This section is wholly devoted to the enunciation and proof and numerical exemplification of propo- sitions concerning the divisions of figures. Favaro reproduces the enunciations of the propositions and numbers them 1 to 5742. He points out that in both enunciation and proof Leonardo 3, 10, 51, 57 are identical with Woepcke 19, 20, 29, 28 respectively. But considerably more remains to be remarked. 11. No less than twenty-two of Woepcke’s propositions are practically iden- tical in statement with propositions in Leonardo; the solutions of eight more of Woepcke are either given or clearly indicated by Leonardo’s methods, and all six of the remaining Woepcke propositions (which are auxiliary) are assumed as known in the proofs which Leonardo gives of propositions in Woepcke. Indeed, these two works have a remarkable similarity. Not only are practically all of the Woepcke propositions in Leonardo, but the proofs called for by the order of 38 L. F. Ofterdinger, Beiträge zur Wiederherstellung der Schrift des Euklides über der Theilung der Figuren, Ulm, 1853. 39 M. Cantor, Vorlesungen über Geschichte der Mathematik, ii2, 1900, pp. 3–53; “Practica Geometriae,” pp. 35–40. 40 Scritti di Leonardo Pisano matematico del secolo decimoterzo pubblicati da Bal- dassarre Boncompagni. Volume ii (Leonardi Pisani Practica Geometriae ed opuscoli). Roma...1862. Practica Geometriae, pp. 1–224. 41 Scritti di Leonardo Pisano...ii, pp. 110–148. 42 These numbers I shall use in what follows. Favaro omits some auxiliary proposi- tions and makes slips in

connection with 28 and 40. Either 28 should have been more general in statement or another number should have been introduced. Similarly for 40. Compare Articles 33–34, 35.

10 EUCLID’S BOOK ON DIVISION OF FIGURES I [12–13 the propositions and by the auxiliary propositions in Woepcke are, with a pos- sible single exception91, invariably the kind of proofs which Euclid might have given—no other propositions but those which had gone before or which were to be found in theElements being required in the successive constructions. Leonardo had a wide range of knowledge concerning Arabian mathematics and the mathematics of antiquity. HisPractica Geometriaecontains many ref- erences to Euclid’s Elements and many uncredited extracts from this work43. Similar treatment is accorded works of other writers. But in the great elegance, finish and rigour of the whole, originality of treatment is not infrequently evi- dent. If Gherard of Cremona made a translation of Euclid’s bookOn Divisions, it is not at all impossible that this may have been used by Leonardo. At any rate the conclusion seems inevitable that he must have had access to some such MS. of Greek or Arabian origin. Further evidence that Leonardo’s work was of Greek-Arabic extraction can be found in the fact that, in connection with the 113 figures, of the section On Divisions, of Leonardo’s work, the lettering in only 58 contains the lettersc or f; that is, the Greek-Arabic successiona, b, g, d, e, z ... is used almost as frequently as the Latina, b, c, d, e, f, g,...; elimination of Latin letters added to a Greek succession in a figure, for the purpose of numerical examples (in which the work abounds), makes the balance equal. 12. My method of restoration of Euclid’s work has been as follows. Every- thing in Woepcke’s text (together with his notes) has been translated literally, reproduced without change and enclosed by quotation marks. To all of Euclid’s enunciations (unaccompanied by constructions) which corresponded to enun- ciations by Leonardo, I have reproduced Leonardo’s constructions and proofs, with the same lettering of the figures44, but occasional abbreviation in the form of statement; that is, the extended form of Euclid in Woepcke’s text, which is also employed by Leonardo, has been sometimes abridged by modern notation or briefer statement. Occasionally some very obvious steps taken by Leonardo have been left out but all such places are clearly indicated by explanation in square brackets, [ ]. Unless stated to the contrary, and indicated by differ- ent type, no step is given in a construction or proof which is not contained in Leonardo. When there is no correspondence between Woepcke and Leonardo I have exercised care to reproduce Leonardo’s methods in other propositions, as closely as possible. If, in a given proposition, the method is extremely obvious on account of what has gone before, I have sometimes given little more than an indication of the propositions containing the essence of the required construc- tion and proof. In the case of the six auxiliary propositions, the proofs supplied seemed to be readily suggested by propositions in Euclid’sElements. 43 For example, on pages 15–16, 38, 95, 100–1, 154. 44 This is done in order to give indication of the possible origin of the construction in question (Art. 11).

14] SYNOPSIS OF MUHAMMED’S TREATISE 11 13. Immediately after the enunciations of Euclid’s problems follow the statements of the correspondence with Leonardo; if exact, a bracket encloses the number of the Leonardo proposition,

according to Favaro’s numbering, and the page and lines of Boncompagni’s edition where Leonardo enunciates the same proposition. The following is a comparative table of the Euclid and, in brackets, of the corresponding Leonardo problems: 1 (5); 2 (14); 3 (2, 1); 4 (23); 5 (33); 6 (16); 7 (20)45; 8 (27)46; 9 (30, 31)47; 10 (18); 11 (0); 12 (28)42; 13 (32)47; 14 (36); 15 (40); 16 (37); 17 (39); 18 (0); 19 (3); 20 (10); 21 (0); 22 (0); 23 (0); 24 (0); 25 (0); 26 (4); 27 (11); 28 (57); 29 (51)45; 30 (0); 31 (0); 32 (29); 33 (35); 34 (40)42; 35 (0); 36 (0). Summary It will be instructive, as a means of comparison, to set forth in synoptic fashion: (1) the Muhammed-Commandinus treatise; (2) the Euclid treatise; (3) Leonardo’s work. In (1) and (2) I follow Woepcke closely48. 14. Synopsis of Muhammed’s Treatise— I. In all the problems it is required to divide the proposed figure into two parts having a given ratio. II. The figures divided are: the triangle (props. 1–6); the parallelogram (11); the trapezium89 (8, 12, 13); the quadrilateral (7, 9, 14–16); the pentagon (17, 18, 22); a pentagon with two parallel sides (19), a pentagon of which a side is parallel to a diagonal (20). III. The transversal required to be drawn: A. passes through a given point and is situated: 1. at a vertex of the proposed figure (1, 7, 17); 2. on any side (2, 9, 18); 3. on one of the two parallel sides (8). B. is parallel: 1. to a side (not parallel) (3, 13, 14, 22); 2. to the parallel sides (11, 12, 19); 45 Leonardo considers the case of “one third” instead of Euclid’s “a certain fraction,” but in the case of 20 he concludes that in the same way the figure may be divided “into four or many equal parts.”Cf. Article 28. 46 Woepcke 8 may be considered as a part of Leonardo 27 or better as an unnumbered proposition following Leonardo 25. 47 Leonardo’s propositions 30–32 consider somewhat more general problems than Euclid’s 9 and 13.Cf. Articles 30 and 34. 48 Woepcke, pp. 245–246.

12 EUCLID’S BOOK ON DIVISION OF FIGURES I [15–16 3. to a diagonal (15, 20); 4. to a perpendicular drawn from a vertex of the figure to the opposite side (4); 5. to a transversal which passes through a vertex of the figure (5); 6. to any transversal (6, 16). IV. Prop. 10: Being given the segmentAB and two lines which pass through the extremities of this segment and form with the lineAB any angles, draw a line parallel toAB from one or the other side ofAB and such as to produce a trapezium of given size. Prop. 21. Auxiliary theorem regarding the pentagon. 15. Commandinus’s Treatise—Appended to the first published edition of Muhammed’s work was a short treatise49 by Commandinus who said50 of Mu- hammed: “for what things the author of the book hath at large comprehended in many problems, I have compendiously comprised and dispatched in two only.” This statement repeated by Ofterdinger51 and Favaro52 is somewhat misleading. The “two problems” of Commandinus are as follows: “Problem I. To divide a right lined figure according to a proportion given, from a point given in any part of the ambitus or circuit thereof, whether the said point be taken in any angle or side of the figure.” “Problem II. To divide a right lined figureGABC, according to a proportion given,E to F, by a right line parallel to another given lineD.” But the first problem is divided into 18 cases: 4 for the triangle, 6 for the quadrilateral, 4 for the pentagon, 2 for the hexagon and 2 for the heptagon; and the second problem, as Commandinus treats it, has 20 cases: 3 for the triangle, 7 for the quadrilateral, 4 for the pentagon, 4 for the hexagon, 2 for the heptagon. 16. Synopsis of Euclid’s Treatise— I. The proposed figure is divided: 1. into two equal

parts (1, 3, 4, 6, 8, 10, 12, 14, 16, 19, 26, 28); 2. into several equal parts (2, 5, 7, 9, 11, 13, 15, 17, 29); 3. into two parts, in a given ratio (20, 27, 30, 32, 34, 36); 4. into several parts, in a given ratio (31, 33, 35, 36). The construction 1 or 3 is always followed by the construction of 2 or 4, except in the propositions 3, 28, 29. 49 Commandinus11, pp. 54–76. 50 Commandinus11, p. [ii]; Leeke-Serle Euclid, p. 603. 51 Ofterdinger38, p. 11, note. 52 Favaro6, p. 139.

17] ANALYSIS OF LEONARDO'S WORK 13 II. The figures divided are: the triangle (1, 2, 3, 19, 20, 26, 27, 30, 31); the parallelogram (6, 7, 10, 11); the trapezium (4, 5, 8, 9, 12, 13, 32, 33); the quadrilateral (14, 15, 16, 17, 34, 35, 36); a figure bounded by an arc of a circle and two lines (28); the circle (29). III. It is required to draw a transversal: A. passing through a point situated: 1. at a vertex of the figure (14, 15, 34, 35); 2. on any side (3, 6, 7, 16, 17, 36); 3. on one of two parallel sides (8, 9); 4. at the middle of the arc of the circle (28); 5. in the interior of the figure (19, 20); 6. outside the figure (10, 11, 26, 27); 7. in a certain part of the plane of the figure (12, 13) B. parallel to the base of the proposed figure (1, 2, 4, 5, 30–33). C. parallel to one another, the problem is indeterminate (29). IV. Auxiliary propositions: 18. To apply to a given line a rectangle of given size and deficient by a square. 21, 22, whena□d□b□c, it follows thata :b□c :d; 23, 24, whena :b>c :d, it follows that (a∓b) :b> (c∓d) :d; 25, whena :b<c :d, it follows that(a−b) :b< (c−d) :d. In the synopsis of the last five propositions I have changed the original no- tation slightly. 17. Analysis of Leonardo's Work. I have not thought it necessary to intro- duce into this analysis the unnumbered propositions referred to above42. I. The proposed figure is divided: 1. into two equal parts (1–5, 15–18, 23–28, 36–38, 42–46, 53–55, 57); 2. into several equal parts (6, 7, 9, 13, 14, 19, 21, 33, 47–50, 56);

14 EUCLID'S BOOK ON DIVISION OF FIGURES I [17 3. into two parts in a given ratio (8, 10–12, 20, 29–32, 34, 39, 40, 51, 52); 4. into several parts in a given ratio (22, 35, 41). The construction 1 or 3 is always followed by the construction of 2 or 4 except in the propositions 42–46, 51, 54, 57. II. The figures divided are: the triangle (1–14); the parallelogram (15–22); the trapezium (23–35); the quadrilateral (36–41); the pentagon (42–43); the hexagon (44); the circle and semicircle (45–56); a figure bounded by an arc of a circle and two lines (57). III. (i) It is required to draw a transversal: A. passing through a point situated: 1. at a vertex of the figure (1, 6, 26, 31, 34, 36, 41–44); 2. on a side not produced (2, 7, 8, 16, 20, 37, 39); 3. at a vertex or a point in a side (40); 4. on one of two parallel sides (24, 25, 27, 30); 5. on the middle of the arc of the circle (53, 55, 57); 6. on the circumference or outside of the circle (45); 7. inside of the figure (3, 10, 15, 17, 46); 8. outside of the figure (4, 11, 12, 18); 9. either inside or outside of the figure (38); 10. either inside or outside or on a side of the figure (32); 11. in a certain part of the plane of the figure (28). B. parallel to the base of the proposed figure (5, 14, 19, 21–23, 29, 33, 35, 54); C. parallel to a diameter of the circle (49, 50). (ii) It is required to draw more than one transversal (a) through one point (9, 47, 48, 56); (b) through two points (13); (c) parallel to one another, the problem is indeterminate (51). (iii) It is required to draw a circle (52).

17] ANALYSIS OF LEONARDO'S WORK 15 IV. Auxiliary Propositions: Although not explicitly stated or proved, Leonardo makes use of four out of six of Euclid's auxiliary

propositions113. On the other hand he proves two other propositions which Favaro does not number: (1) Triangles with one angle of the one equal to one angle of the other, are to one another as the rectangle formed by the sides about the one angle is to that formed by the sides about the equal angle in the other; (2) the medians of a triangle meet in a point and trisect one another.

II. 18. Abraham Savasorda, Jordanus Nemorarius, Luca Paciuolo.—In earlier articles (10, 11) incidental reference was made to Leonardo's general indebted- ness to previous writers in preparing hisPractica Geometriae, and also to the debt which later writers owe to Leonardo. Among the former, perhaps mention should be made of Abraham bar Chijja ha Nasi53 of Savasorda and hisLiber em- badorum known through the Latin translation of Plato of Tivoli. Abraham was a learned Jew of Barcelona who probably employed Plato of Tivoli to make the translation of his work from the Hebrew. This translation, completed in 1116, was published by Curtze, from fifteenth century MSS., in 190254. Pages 130–159 of this edition contain "capitulum tertium in arearum divisionum explanatione" with Latin and German text, and among the many other propositions given by Savasorda is that of Proclus-Euclid (= Woepcke 28 = Leonardo 57). Compared with Leonardo's treatment of divisions Savasorda's seems rather trivial. But however great Leonardo's obligations to other writers, his originality and power sufficed to make a comprehensive and unified treatise. Almost contemporary with Leonardo was Jordanus Nemorarius (d. 1237) who was the author of several works, all probably written before 1222. Among these isGeometria vel De Triangulis55 in four books. The second book is prin- cipally devoted to problems on divisions: Propositions 1–7 to the division of lines and Propositions 8, 13, 17, 18, 19 to the division of rectilineal figures. The enunciations of Propositions 8, 13, 17, 19 correspond, respectively, to Euclid 3, 26, 19, 14 and to Leonardo 2, 4, 3, 36. But Jordanus's proofs are quite dif- ferently stated from those of Euclid or Leonardo. Both for themselves and for comparison with the Euclidean proofs which have come down to us, it will be interesting to reproduce propositions 13 and 17 of Jordanus. "13. Triangulo dato et puncto extra ipsum signato lineam per punctum transeuntem designare, que triangulum per equalia parciatur" [pp. 15–16]. 53 That is, Abraham son of Chijja the prince. Cf. Steinschneider, Bibliotheca Mathematica, 1896, (2),x, 34–38, andCantor, Vorlesungen über Geschichte d. Math. i3, 797–800, 907. 54 M. Curtze, "Urkunden zur Geschichte der Mathematik im Mittelalter und der Renaissance..." Erster Teil (Abhandlung zur Geschichte der Mathematischen Wissen- schaften... XII. Heft), Leipzig, 1902, pp. 3–183. 55 Edited with Introduction byMax Curtze, Mitteilungen des Copernicus–Vereins für Wissenschaften und Kunst zu Thorn. vi. Heft, 1887. In his discussion of the second book, Cantor (Vorlesungen ü. Gesch. d. Math.ii2, 75) is misleading and inaccurate. One phase of his inaccuracy has been referred to byEneström (Bibliotheca Mathemat- ica, Januar, 1912, (3),xii, 62).

18 EUCLID'S BOOK ON DIVISION OF FIGURES II [18 a b c d e f g h k l m n p "Let abc be the triangle andd the point outside but contained within the lines aef, hbl, which are lines dividing the triangle equally and produced. For if d be taken in any such place, drawdg parallel to ca meeting cb produced ing. Join cd and find mn such that △cdg :△aec (= 1 2△abc) = cg : mn. Then divide cg in k such that gk : kc = kc : mn. Produce dk to meet ca in p. Then I say that dp divides the triangle abc into equal

parts. For, since the trianglеckp is similar to the trianglekdg, by 4 of sixth56 and parallel lines and 15 of first and definitions of similar areas, △ckp :△kdg = mn : kg by corollary to 17 of sixth57. But △kdg :△cdg = kg : cg. Therefore, by equal proportions, △ckp :△cdg = mn : cg. □△ckp :△cdg =△aec :△cdg. And △ckp =△aec (= 1 2△abc) 56 That is, Euclid'sElements, vi. 4. 57 I do not know the MS. of Euclid here referred to; but manifestly it is the Porism of Elements vi. 19 which is quoted: "If three straight lines be proportional, then as the first is to the third, so is the figure described on the first to that which is similar and similarly described on the second."

18] JORDANUS NEMORARIUS 19 by 9 of fifth, and this is the proposition. And by the same process of deduction we may be led to an absurdity, namely, that all may equal a part if the pointk be otherwise than betweene and b or the pointp be otherwise than betweenh anda; the part cut off must always be either all or part of the triangleaec." "17. Puncto infra propositum trigonum dato lineam per ipsum deducere, que triangulum secet per equalia" [pp. 17–18]. a b c d e f g h k m n t y z "Let abc be the triangle andd the point inside and contained within the part between ag and be which divide two sides and triangle into equal parts. Throughd draw fdh parallel toac and drawdb. Then by 12 of this book58 draw mn such that bf : mn =△bdf :△bec (= 1 2△abc). Also find ty such that bf : ty =△bfh :△bec. And since △bfh >△bdf, mn > ty by 8 and 10 of fifth. Now bf : bc = bc : ty 58 That is,De Triangulis, Book 2, Prop. 12: "Data recta linea aliam rectam inuenire, adquamsehabeatpriorsicutquilibetdatustriangulusadquemlibetdatumtriangulum" [p. 15].

20 EUCLID'S BOOK ON DIVISION OF FIGURES II [18 by corollary to 17 of sixth581, and△bfh <△bec since fh, ce are parallel lines. But bc : ty > bc : mn by second part of 8 of fifth. □ bf : bc> bc : mn; □ fc< 1 4mn by 6 of this book59. Add then to the linecf, fromf, a linefz, by 5 of this book60, such that fz : zc = zc : mn; and fz will be less thanfb by the first part of the premise. [Supposition with regard tod?] Join zd and produce it to meetac in k; then I say that the linezdk divides the triangle abc into equal parts. For △bdf :△zdf = bf : zf by 1 of sixth. But △zdf :△zkc = zf : mn by corollary to 17 of sixth57 and similar triangles. Therefore by 1 and by equal proportions △bdf :△zkc = bf : mn. But △bdf :△bec = bf : mn. Therefore by the second part of 9 of fifth △zkc =△bec = 1 2△abc." q.e.f. 581 Rather is it the converse of this corollary, which is quoted in note 57. It follows at once, however: bf : ty =△bfh :△bec = bf 2 : bc2, □ bf□ ty = bc2 or bf : bc = bc : ty. 59 "Cum sit linee breuiori adiecte major proporcio ad compositam, quam composite ad longiorem, breuiorem quarta longioris minorem esse necesse est [p. 13]. 60 "Duabus lineis propositis, quarum una sit minor quarta alterius uel equalis, mi- nori talem lineam adiungere, ut, que adiecte ad compositam, eadem sit composite ad reliquam propositarum proporcio" [p. 12].

19] ARABIAN WRITERS ON DIVISIONS OF FIGURES 21 Proposition 18 of Jordanus is devoted to finding the centre of gravity of a triangle601 anditisstatedintheformofaproblemondivisions. InLeonardothis problem is treated109 by showing that the medians of a triangle are concurrent; but in Jordanus (as in Heron83) the question discussed is, "to find a point in a triangle such that when it is joined to the angular points, the triangle will be divided into three equal parts"(p. 18).

A much later work,Summa de Arithmetica Geometria Proportioni et Proportionalita... by Luca Paciuolo (b. about 1445) was published at Venice in 149461. In the geometrical section (the second, and separately paged) of the work, pages 35 verso–43 verso, problems on divisions of figures are solved, and in this con- nection the author acknowledges great debt to Leonardo's work. Although the treatment is not as full as Leonardo's, yet practically the same figures are em- ployed. The Proclus-Euclid propositions which have to do with the division of a circle are to be found here. 19. "Muhammed Bagdedinus" and other Arabian writers on Divisions of Figures.—We have not considered so far who "Muhammed Bagdedinus" was, other than to quote the statement of Dee35 that he may have been "thatAl- bategnus whom Copernicus often cites as a very considerable author, or that Machomet who is said to have been Alkindus's scholar." Albategnius or Mu- hammed b. Gâbir b. Sinân, Abû 'Abdallâh, el Battânî who received his name from Battân, in Syria, where he was born, lived in the latterpart of the ninth and in the early part of the tenth century62. El-Kindî (d. about 873) the philosopher of the Arabians was in his prime about 85063. "Alkindus's scholar" would there- fore possibly be a contemporary of Albategnius. It is probably because of these suggestions of Dee64 that Chasles speaks65 of "Mahomet Bagdadin, géomètre du xe siècle." Itwouldbescarcelyprofitabletodomorethangivereferencestotherecorded 601 Archimedes proved (Works of Archimedes, Heath ed., 1897, p. 201;Opera omnia iterum edidit J. L. Heiberg,ii, 150–159, 1913) in Propositions 13–14, Booki of "On the Equilibrium of Planes" thatthe centre of gravity of any triangle is at the intersec- tion of the lines drawn from any two angles to the middle points of the opposite sides respectively. 61 A new edition appeared at Toscolano in 1523, and in the section which we are discussing there does not appear to be any material change. 62 M. Cantor, Vorlesungen ü. Gesch. d. Math. i3, 736. 63 M. Cantor, Vorlesungen ü. Gesch. d. Math. i3, 718. 64 Cf. Steinschneider12. 65 Chasles, Aperçu historique... 3e éd., Paris, 1889, p. 497.

22 EUCLID'S BOOK ON DIVISION OF FIGURES II [19 opinions of other writers such as Smith66, Kästner67, Fabricius68, Heilbronner69, Montucla70, Hankel71, Grunert72— whose results Favaro summarizes73. The latest and most trustworthy research in this connection seems to be due to Suter who first surmised74 that the author of the Dee book On Divisions was Muh.. b. Muh.. el-Ba˙ gdâdî who wrote at Cairo a table of sines for every minute. A little later75, however, Suter discovered facts which led him to believe that the true author was Abû Muh.ammed b. 'Abdelbâqî el-Ba˙ gdâdî (d. 1141 at the age of over 70 years) to whom an excellent commentary on Bookx of theElements has been ascribed. Of a MS. by this author Gherard of Cremona (1114–1187) may well have been a translator. Euclid's book On Divisions was undoubtedly the ultimate basis of all Ara- bian works on the same subject. We have record of two or three other treatises. 1. T¯âbit b. Qorra (826–901) translated parts of the works of Archimedes and Apollonius, revised Ishâq's translation of Euclid'sElements and Data and also revised the workOn Divisions of Figurestranslated by an anonymous writer76. 2. Abû Muh.. el-Hasan b. 'Obeidallâh b. Soleimân b. Wahb (d. 901) was a distinguished geometer who wrote "A Commentary on the difficult parts of the work of Euclid" and "The Book on Proportion." Suter

thinks77 that another reading is possible in connection with the second title, and that it may refer to Euclid's workOn Divisions. 3. Abû'l Wefâ el-Bûzˇ gânî (940–997) one of the greatest of Arabian mathe- maticians and astronomers spent his later life in Bagdad, and is the author of a course of Lectures on geometrical constructions. Chaptersvii–ix of the Persian form of this treatise which has come down to us in roundabout fashion were entitled: "On the division of triangles," "On the division of quadrilaterals," "On 66 T. Smith, Vitae quorumdam...virorum, 1707, p. 56.Cf. notes 14, 15. 67 A. G. Kästner, Geschichte der Mathematik..., Bandi, Göttingen, 1796, p. 273. See also his preface toN. Morville, Lehre von der geometrischen und ökonomischen Vertheilung der Felder, nach der dänischen Schrift bearbeitet von J. W. Christiani, begleitet mit einer Vorrede...von A. G. Kästner, Göttingen, 1793. 68 J. A. Fabricius, Bibliotheca Graeca... Editio nova. Volumen quartum, Ham- burgi, mdcclxxxv, p. 81. 69 J. C. Heilbronner, Historia Matheseos universae... Lipsiae, mdccxlii, p. 438, 163–4. 70 J. F. Montucla, Histoire des mathématiques...éd. nouv. Tomei, Anvii, p. 216. 71 H. Hankel, Zur Geschichte der Math. in Alterthum u. Mittelalter, Leipzig, 1874, p. 234. 72 J. A. Grunert, Math. Wörterbuch ...von G.S. Klügel, fortgesetzt von C. B. Mollweide und beendigt von J.A. Grunert...Erste Abteilung, die reine Math., fünfter Theil, erster Band, Leipzig, 1831, p. 76. 73 Favaro, pp. 141–144. 74 H. Suter, "Die Mathematiker und Astronomen der Araber und ihre Werke" (Abh. z. Gesch. d. Math. Wiss.x. Heft, Leipzig, 1900), p. 202, No. 517. 75 H. Suter, idem, "Nachträge und Berichtigung" (Abh. z. Gesch. d. Math. Wiss. xiv. Heft, 1902), p. 181; alsoBibliotheca Mathematica, iv3, 1903, pp. 22–27. 76 H. Suter, "Die Mathematiker...," pp. 34–38. 77 H. Suter, "Die Mathematiker...," pp. 48 and 211, note 23.

20] HERON OF ALEXANDRIA, APOLLONIUS OF PERGA 23 the division of circles" respectively. Chaptervii and the beginning of Chapter viii are, however, missing from the Bibliothèque nationale Persian MS. which has been described by Woepcke78. This MS., which gives constructions without demonstrations, was made from an Arabian text, by one Abû Ishâq b. 'Abdal- lâh with the assistance of four pupils and the aid of another translation. The Arabian text was an abridgment of Abû'l Wefâ's lectures prepared by a gifted disciple. The three propositions of Chapterix79 are practically identical with Euclid (Woepcke) 28, 29. In Chapterviii80 there are 24 propositions. About a score are given, in substance, by both Leonardo and Euclid. In conclusion, it may be remarked that in Chapterxii of Abû'l Wefâ's work are9propositions, withvarious solutions, fordividingthesurfaceofa sphereinto equiangular and equilateral triangles, quadrilaterals, pentagons and hexagons. 20. Practical applications of the problems On Divisions of Figures; the μετρικά of Heron of Alexandria.—The popularity of the problems of Euclid's book On Divisions among Arabians, as well as later in Europe, was no doubt largely due to the possible practical application of the problems in the division of parcels of land of various shapes, the areas of which, according to the Rhind papyrus, were already discussed in empirical fashion about 1800b. c. In the first century before Christ81 we find that Heron of Alexandria dealt with the division of surfaces and solids in the third book of hisSurveying (μετρικά)82. Although the enunciations of the propositions in this book are, as a whole, similar83 to those in Euclid's book On Divisions, Heron's discussion

consists 78 F. Woepcke, “Recherches sur l’histoire des Sciences mathématiques chez les orientaux, d’après des traités inedits Arabes et Persans. Deuxième article. Analyse et extrait d’un recueil de constructions géométriques par Aboûl Wafâ,”Journal asiatique, Fevrier–Avril, 1855, (5), V, 218–256, 309–359; reprint, Paris, 1855, pp. 89. 79 F. Woepcke, idem, pp. 340–341; reprint, pp. 70–71. 80 F. Woepcke, idem, pp. 338–340; reprint, pp. 68–70. 81 This date is uncertain, but recent research appears to place it not earlier than 50 b. c. nor later than 150a. d. Cf. Heath, Thirteen Books of Euclid’s Elements, i, 20–21; or perhaps better still, Article “Heron 5” by K. Tittel in Pauly-Wissowa’s Real-Encyclopädie der class. Altertumswissenschaften, viii, Stuttgart, 1913, especially columns 996–1000. 82 Heronis Alexandrini opera quae supersunt omnia, Vol. iii, Rationes Dimetiendi et commentatio Dioptrica recensuit Hermannus Schoene, Lipsiae,mcmiii. Third book, pp. 140–185. Cf. Cantor, Vorlesungen... , i3, 380–382. 83 Only two are exactly the same:ii–iii (= Euclid 30),vii (= Euclid 32), the prob- lem considered in x is practically Euclid 27 (Art. 48), whilexviii is closely related to Euclid 29 (Art. 50). Inxix Heron finds in a triangle a point such that when it is joined to the angular points, the triangle will be divided into three equal parts. The divisions of solids of which Heron treats are of a sphere (xxiii) and the division in a given ratio, by a plane parallel to the base, of a Pyramid (xx) and of a Cone (xxi). For proof of Propositionxxiii: To cut a sphere by a plane so that the volumes of the segments are to one another in a given ratio, Heron refers to Proposition 4, Bookii of “On the Sphere and Cylinder” of Archimedes; the third proposition in the same book

24 EUCLID’S BOOK ON DIVISION OF FIGURES II [21 almost entirely of “analyses” and approximations. For example,ii: “To divide a triangle in a given ratio by a line drawn parallel to the base”—while Euclid gives thegeneralconstruction, Heronconsidersthatthesidesofthegiventrianglehave certain known numerical lengths and thence finds the approximate distance of the angular points of the triangle to the points in the sides where the required line parallel to the base intersects them, because, as he expressly states, in a field with uneven surface it is difficult to draw a line parallel to another. Most of the problems are discussed with a variety of numbers although theoretical analysis sometimes enters. Take as an example Propositionx84: “To divide a triangle in a given ratio by a line drawn from a point in a side produced.” A B Z E Γ Δ “Suppose the construction made. Then the ratio of triangleAEZ to quadri-lateral ZEBΓ is known; also the ratio of the triangleABΓ to the triangleAZE. But the triangle ABΓ is known, therefore so is the triangleAZE. Now Δ is given. Through a known point Δ there is therefore drawn a line which, with two linesAB and AΓ intersecting inA, encloses a known area. Therefore the pointsE and Z are given. This is shown in the second book of On Cutting off a Space. Hence the required proof. If the pointΔ be not onBΓ but anywhere this will make no difference.” 21. Connection between Euclid’s book On Divisions, Apollonius’s treatise On Cutting off a Space and a Pappus-lemma to Euclid’s book of Porisms.— Although the name of the author of the above-mentioned work is not given by of the Archimedean work is (Heronxvii): To cut a given sphere by a plane so that the surfaces of the segments may have to one another a given ratio. (Works of Archimedes, Heath ed., 1897, pp. 61–65;Opera omniaiterum edidit J. L. Heiberg,i, 184–195, 1910.) Propositions ii and vii are also

given in Heron’sπερὶ διόπτρας (Schoene’s edition, pp. 278–281). Cf. “Extraits des Manuscrits relatifs à la géométrie grecs” par A. J. C. Vincent, Notices et extraits des Manuscrits de la bibliothèque impériale, Paris, 1858, xix, pp. 157, 283, 285. 84 Heron, idem, p. 160f.

21] HERON OF ALEXANDRIA, APOLLONIUS OF PERGA 25 Heron, the reference is clearly to Apollonius’s lost work. According to Pappus it consisted of two books which contained 124 propositions treating of the various cases of the following problem:Given two coplanar straight linesA1 P1, B2 P2, on whichA1 and B2 are fixed points; it is required to draw through a fixed point Δ of the plane, a transversalΔZE forming on A1 P1, B2 P2 the two segments A1 Z, B2 E such that A1 Z□ B2 E is equal to a given rectangle. Given a construction for the particular case whenA1 P1, B2 P2 meet inA, and whenA1 and B2 coincide withA—Heron’s reasoning becomes clear. The solution of this particular case is practically equivalent to the solution of Euclid’s Proposition 19 or 20 or 26 or 27. References to restorations of Apollonius’s work are given in note 111. To complete the list of references to writers before 1500, who have treated A B D E H ΓZ of Euclid’s problems here under discussion, I should not fail to mention the last of the 38 lemmas which Pappus gives as useful in connection with the 171 theorems of Euclid’s lost book of Porisms: Through a given point E in BD produced to draw a line cutting the parallelogramAD such that the triangleZΓH is equal to the parallelogramAD. After “Analysis” Pappus has the following “Synthesis. Given the parallelogram AD and the point E. Through E draw the line EZ such that the rectangle ΓZ□ ΓH equals twice the rectangle AΓ□ ΓD. Then according to the above analysis [which contains a reference to an earlier lemma discussed a little later88 in this book] the triangleZΓH equals the parallelogram AD. Hence EZ satisfies the problem and is the only line to do so85.” The tacit assumption here made, that the equivalent of a proposition of Euclid’s bookOn Divisions (of Figures)was well known, is noteworthy. 85 Pappus ed. by Hultsch, Vol. 2, Berlin, 1877, pp. 917–919. In Chasles’s restoration of Euclid’s Porisms, this lemma is used in connection with “Porism clxxx: Given two lines SA, SA′, a pointP and a spaceν: points I and J′ can be found in a line with P and such that if one take onSA, SA′ two pointsm, m′, bound by the equation Im□J′m′ =ν, the linemm′ will pass through a given point.”Les trois livres de Porismes d’Euclide, Paris, 1860, p. 284. See also the restoration by R. Simson, pp. 527–530 of “De porismatibus tractatus,”Opera quaedam reliqua... Glasguae, m.dcc.lxxvi.

III. “The Treatise of Euclid on the Division(of plane Figures).” Proposition 1. 22. “To divide86 a given triangle into equal parts by a line parallel to its base.” [Leonardo 5, p. 119, ll. 7–9.] Let abg be the given triangle which it is required to bisect by a line parallel to bg. Produce ba to d till ba = 2ad. Then inba find a pointe such that ba : ae = ae : ad. Through e draw ez parallel to bg; then the triangle abg is divided by the line ez into two equal parts, of which one is the triangleaez, and the other the quadrilateral ebgz. a b d e g z Leonardo then gives three proofs, but as the first and second are practically equivalent, I shall only indicate the second and third. I. When three lines are proportional, as the first is to the third so is a figure on the first to the similar and similarly situated figure described on the second [vi. 19, “Porism”]87. □ ba : ad = figure on ba : similar and similarly situated figure onae. 86 Literally, the

original runs, according to Woepcke, “We propose to ourselves to demonstrate how to divide, etc.” I have added all footnotes except those attributed to Woepcke. 87 Throughout the restoration I have added occasional references of this kind to Heath’s edition of Euclid’sElements; vi. 19 refers to Proposition 19 of Bookvi. Cf. note 57.

28 EUCLID’S BOOK ON DIVISION OF FIGURES III [22 Hence ba : ad =△abg :△aez = 2 : 1 . □△abg = 2△aez. II. ba : ae = ae : ad. □ ba□ ad = ae2, and since ad is one-half ofba, ba2 = 2 ae2. And since bg is parallel toez, ba : ae = ga : az. □ ba2 : ae2 = ga2 : az2. [vi. 22] But ba2 = 2 ae2. □ ga2 = 2 az2. Then ba□ ag = 2 ae□ az, [vi. 22] □△abg = 2△aez 88. Then follows a numerical example. 88 The theorem here assumed is enunciated by Leonardo (p. 111, ll. 24–27) as follows: a b c d e f Et si á trigono recta protracta fuerit secans duo latera trigonj, que cum ipsis duobus lateribus faciant trigonum habentem angulum unum comunem cum ipso trigono, erit proportio unius trigoni ad alium, sicut facta ex lateribus continentibus ipsum angulum. This is followed by the sentence “Ad cuius rei euidentiam.” Then come the construction and proof: Let abc be the given triangle andde the line across it, meeting the sidesca and cb in

22] PROPOSITION 1 29 the pointsd, e, respectively. I say that △abc :△dec = ac□ cb : dc□ ce. Proof: To ac apply the triangleafc =△dec. [i. 44] Since the trianglesabc, afc are of the same altitude, bc : fc =△abc :△afc. [vi. 1] But bc : fc = ac□ bc : ac□ fc, [v. 15] □△abc :△afc = ac□ bc : ac□ fc, and since△dec =△acf, △acb :△dce = ac□ bc : ac□ cf. Again, since the triangles acf, dce are equal and have a common angle, as in the fifteenth theorem of the sixth book of Euclid, the sides are mutually proportional. □ ac : dc = ce : cf, □ ac□ cf = dc□ ce, □△acb :△dce = ac□ cb : dc□ ce. “quod oportebat ostendere.” It is to be observed that the Latin letters are used with the above figure. This suggests the possibility of the proof being due to Leonardo. The theorem is assumed in Euclid’s proof of proposition 19 (Art. 40) and it occurs, directly or indirectly, in more than one of his works. A proof, depending on the proposi- tion that the area of a triangle is equal to one-half the product of its base and altitude, is given by Pappus (pp. 894–897) in connection with one of his lemmas for Euclid’s book ofPorisms: Triangles which have one angle of the one equal or supplementary to one angle of the other are in the ratio compounded of the ratios of the sides about the equal or supplementary angles. (Cf. R. Simson, “De Porismatibus Tractatus” inOpera quaedam reliqua... 1776, p. 515 ff.—P. Breton (de Champ), “Recherches nouvelles sur les porismes d’Euclide,”Journal de mathématiques pures et appliquées, xx, 1855, p. 233 ff. Reprint, p. 25 ff.—M. Chasles, Les trois livres de Porismes d’Euclide... Paris, 1860, pp. 247, 295, 307.) The first part of this lemma is practically equivalent to either (1) [vi. 23]:Equiangular parallelograms have to one another the ratio compounded of the ratio of their sides; or (2) the first part of Prop. 70 of theData (Euclidis Data... edidit H. Menge, Lipsiae, 1896, p. 130f.):If in two equiangular parallelograms the sides containing the equal angles have a given ratio to one another[i. e. one side in one to one side in the other],the parallelograms themselves will also have a given ratio to one another. Cf. Heath, Thirteen Books of Euclid’s Elements, ii, 250. The proposition is stated in another way by Pappus85 (p. 928) who proves that a parallelogram is to an equiangular parallelogram as the rectangle contained by the adjacent sides of the first is to the rectangle contained by the

adjacent sides of the second. The above theorem of Leonardo is precisely the first of those theorems which Com- mandinus adds tovi. 17 ofhis edition of Euclid'sElements and concerning which he writes "à nobis elaborata" ("fatti da noi"):Euclidis Elementorum Libri XV...A Fed- erico Commandino... Pisauri, mdlxxii, p. 81 recto (Degli Elementi d' Euclide libri

30 EUCLID'S BOOK ON DIVISION OF FIGURES III [22–23 Proposition 2. 23. "To divide a given triangle into three equal parts by two lines parallel to its base."[Leonardo 14, p. 122, l. 8.] a b d e g i k tz Let abgbethegiventrianglewithbase bg. Produce bato dtill ba = 3ad, and produce ad to e till ad = de; then ae = 2 3ba. Find az, a mean proportional between ba and ad, and ia a mean proportional betweenba and ae. Then throughz and i draw zt, ik parallel to bg and I say that the triangle abg is divided into three equal parts of which one is the triangle azt, another the quadrilateralzikt, the third the quadrilateralibgk. Proof: Since ba : az = az : ad, ba : ad =△abg :△azt, [vi. 19, Porism] for these triangles are similar. Now ba = 3ad; □△abg = 3△azt. □△azt = 1 3△abg. Again, ba : ia = ia : ae; □ ba : ae =△ on ea: similar and similarly situated△ on ai. But triangles aik, abg are similar and similarly described onai and ab; and ea : ab = 2 : 3 . □△aik = 2 3△abg. quindici con gli scholii antichi tradotti prima in lingua latina daM. Federico Com- mandino da Urbino, et con commentarii illustrati, et hora d' ordine dell' istesso trans- portati nella nostra vulgare, et da lui riveduti.In Urbino,m.d.lxxv, p. 88recto).

23–24] PROPOSITION 3 31 And since △azt = 1 3△abg, there remains the quadrilateral zikt = 1 3△abg. We see that the quadrilateral ibgk will be the other third part; hence the triangleabg has been divided into three equal parts; "quod oportebat facere." Leonardo continues: "Et sic per demonstratos modos omnia genera trigonorum possunt diuidi in quatuor partes uel plures."Cf. note 45. Proposition 3. 24. "To divide a given triangle into two equal parts by a line drawn from a given point situated on one of the sides of the triangle."[Leonardo 1, 2, p. 110, l. 31; p. 111, ll. 41–43.] a b dg Given the trianglebgd; if a be the middle point ofgd the line ba will divide the triangle as required; either because the triangles are on equal bases and of the same altitude [i. 38; Leonardo 1], or because △bgd :△bad = bd□ dg : bd□ da88. Whence △bgd = 2△bad. But if the given point be not the middle point of any side, letabg be the triangle andd the given point nearer tob than tog. Bisect bg at e and draw ad, ae. Through e draw ez parallel to da; join dz. Then the triangle abg is bisected bydz. a b d e g z

32 EUCLID'S BOOK ON DIVISION OF FIGURES III [24–25 Proof: Since ad‖ ez, △adz =△ade. To each add△abd. Then quadl.abdz =△abd +△ade, =△abe. But △abe = 1 2△abg; □ quadl.abdz = 1 2△abg; and the triangle zdg is the other half of the triangleabg. Therefore the triangle abg is divided into two equal parts by the linedz drawn from the point d; "ut oportebat facere." Then follows a numerical example. Proposition 4. 25. "To divide a given trapezium89 into two equal parts by a line parallel to its base."[Leonardo 23, p. 125, ll. 37–38.] Let abgd be the given trapezium with parallel sidesad, bg, ad being the lesser. It is required to bisect the trapezium by a line parallel to the base bg. Let gd, ba, produced, meet in a pointe. Determine z such that ze2 = 1 2(eb2 + ea2).90 Through z draw zi parallel togb. I say that the

trapeziumabgd is divided into two equal parts by the linezi parallel to the basebg. a b d e g i z 89 Here, and in what follows, this word is used to refer to a quadrilateral two of whose sides are parallel. 90 The pointz is easily found by constructions which twice make use ofi. 47.

25–26] PROPOSITION 5 33 Proof: For since 2ze2 = eb2 + ea2, and all the triangles are similar, 2△ezi =△ebg +△ead. [vi. 19] From the triangleebg take away the triangleezi. Then △ezi = quadl.zbgi +△eda. And taking away from the equals the triangleeda, we get quadl.ai = quadl.zg. Therefore the trapezium abgd is divided into two equal parts by the line zi parallel to its base. q. o. f. A numerical example then follows. Proposition 5. 26. “And we divide the given trapezium into three equal parts as we divide the triangle, by a construction analogous to the preceding construction91.” [Leonardo 33, p. 134, ll. 14–15.] Let abgd be the trapezium with parallel sides ad, bg and other sides ba, gd produced to meet ine. Let zti be a line such that zi : it = eb2 : ea2.92 91 It is to be noticed that Leonardo’s discussion of this proposition is hardly “analo- gous to the preceding construction” which is certainly simpler than if it had been similar to that of Prop. 5. A construction for Prop. 4 along the same lines, which may well have been Euclid’s method, would obviously be as follows: Let zti be a line such that zi : it =eb2 :ea2. Divide tz into two equal parts,tk, kz. Findm such that em2 :eb2 =ki :zi. Then m leads to the same solution as before. [For, in brief, em2 =eb2 (ki zi) =eb2 (zt 2 +ti zi) = eb2 2 (zi +it zi) = eb2 2 (ea2 +eb2 eb2) = 1 2 (ea2 +eb2) .] 92 Fromvi. 19, Porism, it is clear that the construction here is to find a linex which is a third proportional toeb and ea. Thenzi :it =eb :x.

34 EUCLID’S BOOK ON DIVISION OF FIGURES III [26 Divide tz into three equal partstk, kl, lz. Find m and n in be such that em2 : eb2 = ik : zi, and en2 : eb2 = il : zi. Through m and n draw mo, np parallel to the basebg. Then I say that the quadrilateral ag is divided into three equal parts:ao, mp, ng. Proof: For eb2 : ea2 =△ebg :△ead. [vi. 19] □ zi : it =△ebg :△ead. .....[1] a b d e g i k lm n o p t z But zi : ik = eb2 : em2, □ zi : ik =△ebg :△emo. ...[2] So also zi : il =△ebg :△enp. ...[3] Whence it : tk =△ead : quadl.ao, 93 and therefore tk : kl = quadl.ao : quadl.mp.94 But tk = kl. □ quadl.ao = quadl.mp. So also kl : lz = quadl.mp : quadl.ng; and kl = lz. □ quadl.mp = quadl.ng. 93 This may be obtained by combining [1] and [2], and applyingv. 11, 16, 17. 94 Relations [1], [2] and [3] may be employed, as in the preceding, to give, it : kl =△ead : quadl.mp; combining this withit : tk =△ead : quadl.ao, we get the required result, tk : kl = quadl.ao : quadl.mp.

26–27] PROPOSITION 6 35 Thereforethequadrilateralisdividedintoequalquadrilaterals ao, mp, ng; “ut prediximus.” Then follows a numerical example. Proposition 6. 27. “To divide a parallelogram into two equal parts by a straight line drawn from a given point situated on one of the sides of the parallelogram.” [Leonardo 16, p. 123, ll. 30–31.] Let abcd be the parallelogram andi any point in the sidead. Bisect ad a b c d e f h i k in f and bc in e. Join fe. Then the parallelogramac is divided into equal parallelograms ae, fc on equal bases. Cut off eh = fi. Join hi. Then this is the line required. Leonardo gives two proofs: I. Let hi meet fe in k. Then [△s fki, hke are equal; add to each the pentagon kfabh, etc.] II. Since ae, fc are s, af = be and fd = ec. But fd = 1 2ad. □ fd =

af = ec. And since fi = he, ai = ch. So also di = bh, and hi is common. □ quadl.iabh = quadl.ihcd.95 95 The first rather than the second proof is Euclidean. There is no proposition of the Elements with regard to the equality of quadrilaterals whose sides and angles, taken in the same order, are equal. Of course the result is readily deduced fromi. 4, if we make certain suppositions with regard to order. Cf. the proof of Prop. 10.

36 EUCLID'S BOOK ON DIVISION OF FIGURES III [27–28 Similarly if the given point were betweena and f, [etc.; or on any other side]. And thus a parallelogram can be divided into two equal parts by a straight line drawn from a given point situated on any one of its sides. Proposition 7. 28. "To cut off a certain fraction from a given parallelogram by a straight line drawn from a given point situated on one of the sides of the parallelogram." [Leonardo 20 (the case where the fraction is one-third), p. 124, ll. 24–26.] Let abcd be the given parallelogram. Suppose it be required to cut off a third of this parallelogram, by a straight line drawn fromi, in the side ad. a b c de f g h i k (The figure here is a combination of two in the original.) Trisect ad in e and f and through e, f draw eg, fh parallel to dc; [then these lines trisect the . If the pointi be in the linead, at eithere or f, then the problem is solved. But if it be betweena and e, drawik to bisect the ah (Prop. 6), etc. Similarly ifi were between e and f, or between f and d]. After finishing these cases Leonardo concludes: "eodem modo potest omne paralilogramum diuidi in quatuor uel plures partes equales45." The construction in this proposition is limited to the case where "a certain fraction" is the reciprocal of an integer. But more generally, if the fraction were m : n (the ratio of the lengths of two given lines), we could proceed in a very similar way: Divide ad in e, internally, so thatae : ed = m : n− m (n > m). In ad cut offef = ae and throughf draw fh parallel toab. Then, as before, the problem is reduced to Proposition 6. If the pointe should fall ati or in the intervalai the part cut off from the parallelogram by the required line would be in the form of a triangle which might be determined byi. 44.

28–29] PROPOSITION 8 37 Proposition 8. 29. "To divide a given trapezium into two equal parts by a straight line drawn from a given point situated on the longer of the sides of the trapezium." [Part of Leonardo 2796, p. 127, ll. 2–3.] This enunciation means, apparently, "from a given point situated on the longer of the [parallel] sides." At any rate Leonardo gives constructions for the cases when the given point is on any side. These I shall take up successively. The figure is made from more than one of Leonardo's, and there is a slight change in the lettering. a b c d e f g h i k l m no p q r s t v z Let ad be the shorter of the parallel sidesad, bg, which are bisected in t and k respectively. Join tk. Then if bt, gt be joined, [it is clear, from triangles on equal bases and between the same parallels, thattk bisects the trapezium]. [This is Leonardo 24, p. 126, l. 31.] Next consider the given point as any point on the shorter side[Leonardo 25, p. 127, ll. 2–3]. First let the point be at the anglea. Cut offkl in kg, equal toat. Join al, meeting tk in m; then the quadrilateral is divided as required byal. For [the trianglesatm, mkl are equal in all respects, etc.]. Similarly if d were the given point; inkb cut off kn equal to td, and dn divides the quadrilateral into two equal parts which is proved as in the preceding case. [Were the given point anywhere betweena and t the other end of the bisecting line would be betweenk and l. Similarly if the given point were between t and d, the corresponding point would be betweenk and n.] 96

Leonardo 27: “Quomodo quadrilatera duorum laterum equidistantium diuidantur á puncto dato super quodlibet latus ipsius” [p. 129, ll. 2–3].Cf. note 46.

38 EUCLID’S BOOK ON DIVISION OF FIGURES III [29 Although not observed by Favaro, Leonardo now considers: If the given point be in the sidebg; either l, or n, or a point betweenl and n, then the above construction is at once applicable. Suppose, however, that the given point were atb or in the segmentbn, at g or in the segmentlg. First consider the given point atb. Join bd and through n draw nc parallel tobd to meetgd in c. Join bc. Then bc bisects the trapezium. For [abnd is half of the trapeziumag, and the trianglebnd equals the trianglebdc etc.]. Similarly from a given point betweenb and n, a line could be drawn meeting gd between c and d, and dividing the quadrilateral into two equal parts. So also fromg a line gf could be drawn [etc.]; and similarly for a given point between g and l. Leonardo then concludes (p. 127, ll. 37–40): “Jam ostensum est quomodo in duo equa quadrilatera duorum equidis- tantium laterum diuidi debeant á linea protracta ab omni dato puncto super lineas equidistantes ipsius; nunc uero ostendamus quomodo diuidan- tur á linea egrediente á dato puncto super reliqua latera.” This is overlooked by Favaro, though implied in his 27 [Leonardo, p. 129, l. 4]. I may add Leonardo’s discussion of the above proposition although it does not seem to be called for by Euclid. a b g d h i k z Let the point be in the sidegd. For g or c or d or any point between c and d the above constructions clearly suffice. Let us, then, now consider the given point h as between c and g. Draw the line iz parallel to gb to bisect the trapezium (Prop. 4). Supposeh were between g and i. Join zh. Through i draw ik parallel to hz, and meetingab in k. (The lettering of the original figure is somewhat changed.) Join hk, then [this is the line required; since △izh =△kzh, etc.]. [Similarly if h were between i and d.] [So also for points on the lineab.]

29–30] PROPOSITION 9 39 Proposition 9. 30. “To cut off a certain fraction from a given trapezium by a straight line drawn from a given point situated on the longer side of the trapezium.” [Leonardo 30, 3197, p. 133, ll. 17–19, 31.] I shall interpret “longer side” as in Proposition 8, and lead up to the con- sideration of any given point onbg after discussing the cases of points on the shorter side ad. a b c d e g i kl n t z (This figure is made from three of Leonardo’s.) Suppose it be required to divide the trapezium in the ratioez : zi 98. Divide ad, bg in the pointst, k, respectively, such that at : td = ez : zi = bk : kg. Join tk. Then by joiningbt and gt [it is easily seen byvi. 1 andv. 12, that the trapezium ag is divided bytk in the ratioez : zi]. If the given point be ata or d, make kl = at and gn = bl. Join al, dn. [Adding the quadrilateral ak to the congruent triangles with equal sides at, kl, we find al divides the trapezium in the required ratio. Then from vi. 1, dn does the same.] As in Proposition 8, for any pointt′ betweena and t, ort and d, we have a corresponding pointk′ betweenl andk orn andk, such that the linet′k′ divides the trapezium in the given ratio. 97 As 30, Favaro quotes, “Per rectam protractam super duo latera equidistantia quadrilaterum abscisum in data aliqua proportione dividere”; as 31: “Divisionem in eadem proportione ab angulis habere.” 98 Here, as well as in 15 and 36, Leonardo introduces the representation of numbers by straight lines, and in considering these lines he invariably writes the word number in connection with them;e. g.‘number ez: number zi,’ not ez : zi. Euclidean MSS. of the Elements, Booksvii to ix, adopt this

same method. In what follows, I shall use the abbreviated form.

40 EUCLID’S BOOK ON DIVISION OF FIGURES III [30–31 If the given point be inbg atl orn or betweenl andn, the above reasoning suffices. Suppose however that the given point were atb. Join bd. Through n draw nc parallel to bd. Join bc. Then bc divides the trapezium in the required ratio. Similarly for the pointg and for any point betweenb and n, or betweeng and l. Some of the parts which I have filled in above are covered by the general final statement: “nec non et diuidemus ipsum quadrilaterum ab omni puncto dato super aliquod laterum ipsius.....” (Page 134, ll. 10–11. Compare Proposition 13.) Proposition 10. 31. “To divide a parallelogram into two equal parts by a straight line drawn from a given point outside the parallelogram.”[Leonardo 18, p. 124, ll. 5–7.] a b c d e f g k Let abcd be the given parallelogram ande the point outside. Join bd and bisect it ing. Join eg meeting bc in k and produce it to meetad in f. Then the parallelogram has been divided into two equal parts by the line drawn through e, as may be proved by superposition; and one half is the quadrilateral fabk, the other, the quadrilateralfkcd 99. 99 The proof also follows from the equality of the trianglesfgd, bgk, byi. 26 and of the triangles abd, bdc by i. 4. This problem is possible for all positions of the pointe.

31–34] PROPOSITION 11–13 41 Proposition 11. 32. “To cut off a certain fraction from a parallelogram by a straight line drawn from a given point outside of the parallelogram.” This proposition is not explicitly formulated by Leonardo; but the general method he would have employed seems obvious from what has gone before. Suppose it were required to cut off one-third of the given parallelogramac by a line drawn through a pointe outside of the parallelogram. Then by the method of Proposition 7, form a parallelogram two-thirds ofac. There are four such parallelograms with centresg1,g 2,g 3,g 4. Linesl1,l 2,l 3,l 4 through each one of these points ande will bisect a parallelogram (Proposition 10). There are several cases to consider with regard to the position ofe but it may be readily shown that, in one case at least, there is a lineli(i = 1, 2, 3, 4), which will cut off a third of the parallelogramac. Similarly for one-fourth, one-fifth, or any other fraction such asm : n which represents the ratio of lengths of given lines. Proposition 12. 33. “To divide a given trapezium into two equal parts by a straight line drawn from a point which is not situated on the longer side of the trapezium. It is necessary that the point be situated beyond the points of concourse of the two sides of the trapezium.”[Leonardo 28, p. 129, ll. 2–4, and another, unnumbered100.] Proposition 13. 34. “To cut off a certain fraction from a (parallel-) trapezium by a straight line which passes through a given point lying inside or outside the trapezium but so that a straight line can be drawn through it cutting both 100 As Leonardo 28 Favaro gives, “Qualiter quadrilatera duorum laterum equidistan- tium diuidi debeant a dato puncto extra figuram” and entirely ignores the paragraph headed, “De diuisione eiusdem generis, qua quadrilaterorum per rectam transeuntem per punctum datum infra ipsum” [p. 131, ll. 13–14].

42 EUCLID’S BOOK ON DIVISION OF FIGURES III [34 the parallel sides of the trapezium101.” [Part of Leonardo 32 102, p. 134, ll. 11–12.] We first take up Leonardo’s discussion of Proposition 12. In the figure of Proposition 8, supposeal to

be produced in the direc- tions of the pointse and r; tk in the directions ofq and v, dn of z and h, cb of i and o, gf of s and p. Then for [any such exterior pointse, q, z, i, s, r, v, h, o, p, lines are drawn bisecting the trapezium]. If the given point,x, were anywhere in the section of the plane abovead andbetween eaand dz, thelinejoining xto mwould[bythesamereasoning as in Proposition 8] bisect the trapezium. Similarly for all points below nl and between hn and lr [.....so also for all points within the triangles amd, nml]. This seems to be all that Euclid's Proposition 12 calls for. But just as Leonardo considers Proposition 8 for the general case with the given point any- 101 The final clauses of Propositions 12 and 13, in Woepcke's rendering, are the same. I have given a literal translation in Proposition 12. Heath's translation and interpretation (after Woepcke) are given in 13. Concerning 12 and 13 Woepcke adds the following note: "Suppose it were required to cut off thenth part of the trapeziumABDC; make Aα and Cγ respectively equal to thenth parts ofAB and of CD; then AαγC will be the nth part of the trapezium, forγα produced will pass through the intersection ofCA, DB produced. Now to draw through a given pointE the transversal which cuts off a certain fraction of the trapezium, join the middle pointμ of the segmentαγ, and the pointE, by a line; this lineEFG will be the transversal required to be drawn, since the triangle αFμ equals the triangleγGμ. A B C D E F E'F'E'' G' G α μ γ "But when the given point is situated asE′ or E′′ such that the transversal drawn throughμ no longer meets the two parallel sides but one of the parallel sides and one of the two other sides, or the other two sides; then the construction indicated is not valid since CG′μγ is not equal toBF′μα. It appears that this is the idea which the text is intended to express. The 'points of concourse' are the vertices where a parallel side and one of the two other sides intersect; and the expression 'beyond' refers to the movement of the transversal represented as turning about the pointμ." 102 "Quadrilaterum [trapezium] ab omni puncto dato super aliquod laterum ipsius, et etiam ab omni puncto dato infra, uel extra diuidere in aliqua data proportioni."

34–35] PROPOSITION 14 43 where on the perimeter of the trapezium, so here, he discusses the constructions for drawing a line from any point inside or outside of a trapezium to divide it into two equal parts. Leonardo does not give any details of the discussion of Euclid's Proposition 13, but after presentation of the cases given in Proposition 9 concludes:"et diuidemus ipsum quadrilaterum ab omni puncto dato super aliquod lat- erum ipsius, et etiam ab omni puncto dato infra uel extra"[Leonardo 32, p. 134, ll. 10–12]. From Leonardo's discussion in Propositions 8, 9, 12, not only are the neces- sary steps for the construction of 13 (indicated in the Woepcke note above101) evident, but also those for the more general cases, not considered by Euclid, where restrictions are not imposed on the position of the given point. Proposition 14. 35. "To divide a given quadrilateral into two equal parts by a straight line drawn from a given vertex of the quadrilateral."[Leonardo 36, p. 138, ll. 10–11.] Let abcdbe the quadrilateral anda the given vertex. Draw the diagonal bd, meeting the diagonal ac in e. If be, ed are equal, [ac divides the quadrilateral as required]. If be be not equal toed, make bz = zd. Draw zi‖ ac to meet dc in i. Join ai. Then the quadrilateral abcd is divided as required by the lineai. a b c d e i z Proof: Join az and zc. Then the triangles abz, azd are respectively equal to the trianglescbz, cdz. Therefore the quadrilateralabcz is one-half the quadrilateralabcd. And since the

trianglesazc, aic are on the same base and between the same parallels ac, zi, they are equal. To each add the triangleabc.

44 EUCLID'S BOOK ON DIVISION OF FIGURES III [35–36 Then the quadrilateralabcz is equal to the quadrilateralabci. But the quadrilateral abcz is one-half of the quadrilateralabcd. Therefore abci is one-half of the quadrilateralabcd; "ut oportet." Proposition 15. 36. "To cut off a certain fraction from a given quadrilateral by a line drawn from a given vertex of the quadrilateral." [Leonardo 40, p. 140, ll. 36–37.] Let the given fraction be asez : zi, and let the quadrilateral beabcd and the given vertexd. Divide ac in t such that at : tc = ez : zi. If bd pass through t [then bd is the line required]. a b c d e i l t z But if bd do not pass throught it will intersect eitherct or ta; let it intersect ct. Join bt, td. Then quadl.tbcd : quadl.tbad = ct : ta = ez : zi. Draw tl parallel to the diagonalbd, and joindl. Then the quadrilaterals lbcd, tbcd are equal and the construction has been made as required; for ct : ta = ez : zi = quadl.lbcd :△dal. And ifbd intersect ta [a similar construction may be given to divide the given quadrilateral, by a line throughd, into a quadrilateral and triangle in the required ratio]. Leonardo then gives the construction for dividing a quadrilateral in a given ratio by a line drawn through a point which divides a side of the quadrilateral in the given ratio.

36–37] PROPOSITION 16 45 Proposition 16. 37. "To divide a given quadrilateral into two equal parts by a straight line drawn from a given point situated on one of the sides of the quadrilat- eral." [Leonardo 37, p. 138, ll. 28–29.] Let abcd be the given quadrilateral,e the given point. Divide ac into two equal parts by the linedt [Prop. 14]. Joinet. The lineet either is, or is not, parallel todc. a b c d e t z (Two of Leonardo's figures are combined in one, here.) If et be parallel to dc, join ec. Then the quadrilateral ac [is bisected by the lineec, etc.]. If et be not parallel todc, draw dz∥ et. Join ez. Then ac [is bisected by the lineez, etc.]. Leonardo does not consider the case of failure of this construction, namely when dz falls outside the quadrilateral. Suppose in such a case that the problem were solved by a line joininge to a pointz′ (not shown in the figure) ondc. Through t, draw tt′∥ cd. Join ct′. Then △ct′d =△ctd =△edz′. Whence △et′c =△ez′c, ort′z′∥ ce. Therefore fromt′,z′ may be found and the solution in this case is also possible, indeed in more than one way, but it is not in Euclid's manner to consider this question. Should the diagonal db bisect the quadrilateral ac, the discussion is similar to the above. But if the line drawn fromd to bisect the quadrilateral meet the side ab in i, draw bk bisecting the quadrilateralac. If k be not the given point, it will be betweenk and d or between k and a. In the first case joinbe and through k draw kl∥ eb. Join el [then el is the required bisector].

46 EUCLID'S BOOK ON DIVISION OF FIGURES III [37 a b c d e i k l If the pointe be between a and k [a similar construction with the line through k parallel tobe, and meetingbc in m, leads to the solution by the line em]. a b c d e i t z Were e at the middle of a side such asab, draw dz∥ ab and bisect dz in i. Join ei, ci and ec. Through i draw it∥ ec. Join et; then et [bisects the quadrilateral ac, since△itc =△ite, etc.]. If dz were to fall outside the quadrilateral, draw fromc the parallel to ba; and so on.

37–38] PROPOSITION 17 47 Proposition 17. 38. "To cut off a certain fraction from a quadrilateral by a straight line drawn from a given point situated on one of the

sides of the quadrilateral." [Leonardo 39, p. 140, ll. 11–12.] Let abcd be the given quadrilateral and suppose it be required to cut off one-third by a line drawn from the pointe in the sidead. a b c de iz Draw dz cutting off one-third ofac [Prop. 15]. Join ez, ec. If ez‖ dc, then ecd [is the required part cut off, etc.]. But if ez be not parallel todc, draw di‖ ez and join ei. [Then this is the line required, etc.] The case whenei cuts dc is not taken up but it may be considered as in the last proposition. Soalsotodivide ac intoanyratio: draw dz dividingitinthatratio(Prop.15), and then proceed as above. A particular case which Leonardo gives may be added. Let ab be divided into three equal partsae, ef , fb; drawdg‖ ab and cut off gh = 1 3gd. Join fc and through h draw hi‖ fc, meeting dc in i. Join fi; and the quadrilateralfbci will be one-third of the quadrilateralac. [As in latter part of Prop. 16.]

48 EUCLID'S BOOK ON DIVISION OF FIGURES III [38–39 a b c d e f g h i k Then ek may be drawn to bisect the quadrilateralafid [Prop. 16], and thus the quadrilateralabcd will be divided into three equal portions which are the quadrilateralsak, ei, fc. Proposition 18. 39. "To apply to a straight line a rectangle equal to the rectangle con- tained by AB, AC and deficient by a square103." 103 This proposition is interesting as illustrating the method ofapplication of areas which was "one of the most powerful methods on which Greek Geometry relied." The method first appears in theElements ini. 44:To a given straight line to apply, in a given rectilineal angle, a parallelogram equal to a given triangle—a proposition which Heath characterises as "one of the most impressive in all geometry" while the "marvellous ingenuity of the solution is indeed worthy of the 'godlike men of old' as Proclus calls the discoverers of the method of 'application of areas'; and there would seem to be no reason to doubt that the particular solution, like the whole theory, was Pythagorean, and not a new solution due to Euclid himself." [I continue to quote mainly from Heath who may be consulted for much greater detail: Heath, Thirteen Books of Euclid's Elements, i, 9, 36, 343–7, 383–8; ii, 187, 257–67— Heath, Apollonius of Perga Treatise on Conic Sections, Cambridge, 1896, pp. lxxxi–lxxxiv, cii–cxi—Heath, The Works of Archimedes, Cambridge, 1897, pp. xl– xlii, 110 and "Equilibrium of Planes," Bkii, Prop. 1, and "On conoids and spheroids," Props. 2, 25, 26, 29. See also: Cantor, Vorlesungen über Geschichte der Math.i3, 289–291, etc. (under index heading 'Flächenanlegung')—H. G. Zeuthen, Geschichte der Mathematik im Alterthum und Mittelalter, Kopenhagen, 1896, pp. 45–52 (French ed. Paris, 1902, pp. 36–44)— C. Taylor, Geometry of Conics..., Cambridge, 1881, pp. xliii–xliv.] The simple application of a parallelogram of given area to a given straight line as one of its sides is what we have in theElements i. 44 and 45; the general form of the problem with regard toexceedingand falling-short may be stated thus: "To apply to a given straight line a rectangle (or, more generally, a parallelogram) equal to a given rectilineal figure and (1)exceedingor (2)falling-short by a square (or, in the more general case, a parallelogram similar to a given parallelogram)." What is meant by saying that the applied parallelogram (1)exceedsor (2)falls short is that, while its base coincides and is coterminousat one end with the straight line,

39] PROPOSITION 18 49 "After having done what was required, if some one ask, How is it pos- sible to apply to the lineAB a rectangle such that the rectangleAE EB A BCE Z is equal to the rectangleAB AC and deficient by a square—we say that it is

impossible, because AB is greater than BE and AC greater than AE, and consequently the rectangleBA□ AC greater than the rectangle AE□ EB. Then when one applies to the lineAB a parallelogram equal to the rectangle AB□ AC the rectangle AZ□ ZB is..... 104." In this problem it is required to find in the given lineAB a pointZ such that AB□ ZB− ZB2[= AZ□ ZB by ii. 3; cf. x. 16 lemma] = AB□ AC 105. the said base (1) overlaps or (2) falls short of the straight lineat the other end, and the portion by which the applied parallelogram exceeds a parallelogram of the same angle and height on the given straight line (exactly) as base is a parallelogram similar to a given parallelogram (or, in particular cases, a square). In the case where the parallelogram is tofall short, some such remark as Woepcke's (note 104) is necessary to express the condition of possibility of solution. For the other case see note 116. The solution of the problems here stated is equivalent to the solution of a quadratic equation. By means ofii. 5 and 6 we can solve the equations ax±x2 =b2, x2−ax =b2, but invi. 28, 29 Euclid gives the equivalent of the solution of the general equations ax±px2 =A. vi. 28 is: To a given straight line to apply a parallelogram equal to a given rectilineal figure and deficient by a parallelogrammic figure similar to a given one: thus the given rectilineal figure must not be greater than the parallelogram described on the half of the straight line and similar to the defect. The Proposition 18 of Euclid under consideration is a particular case of this problem and as the fragment of the text and Woepcke's note (note 104) are contained in it, doubt may well be entertained as to whether Euclid gave any construction in his book On Divisions. The problem can be solved without the aid of Bookvi of theElements and by means ofii. 5 andii. 14 only, as indicated in the text above. The appropriation of the terms parabola (application), hyperbola (exceeding) and el- lipse (falling-short) to conic sections was first introduced by Apollonius as expressing in each case the fundamental properties of curves as stated by him. This fundamental property is the geometrical equivalent of the Cartesian equation referred to any diam- eter of the conic and the tangent at its extremity as (in general, oblique) axes. More particulars in this connection are given by Heath. The terms "parabolic," "hyperbolic" and "elliptic," introduced by Klein for the three main divisions of Geometry, are appropriate to systems in which a straight angle equals, exceeds and falls short of the angle sum of any triangle.Cf. W. B. Frankland, The First Book of Euclid's Elements with a Commentary based principally upon that of Proclus Diadochus...Cambridge, 1905, p. 122.

50 EUCLID'S BOOK ON DIVISION OF FIGURES III [39 Find, byii. 14, the side,b, of a square equal in area to the rectangleAB□AC, then the problem is exactly equivalent to that of which a simple solution was given by Simson106: To apply a rectangle which shall be equal to a given square, to a given straight line, deficient by a square: but the given square must not be greater than that upon the half of the given line. A B D (F) H N O Z Bisect AB in D, and if the square onAD be equal to the square onb, the thing required is done. But if it be not equal to it,AD must be greater thanb according to the determination. Then drawDO perpendicular to AB and equal to b; produce OD to N so that ON = DB (or 1 2a); and withO as centre and radius ON describe a circle cuttingDB in Z. Then ZB (or x) is found, and therefore the required rectangleAH. For the rectangleAZ□ ZB together with the square onDZ is equal to the square on DB, [ii. 5] i. e. to the square onOZ, i. e. to the squares onOD, DZ. [i. 47] Whence the

rectangleAZ□ ZB is equal to the square onOD. 104 Woepcke here remarks: “Evidently ifa denote the length of the line to which the rectangle is to be applied, Problem 18 is only possible when AB□ AC < (a 2)2 . Then ifa be taken asAB one of the two sides of the given rectangle, relatively to the other side,AC <AB 4 . It is probably the demonstration of this which was given in the missing portion of the text.” 105 If AB =a, ZB =x, AB□ AC =b2, the problem is to find a geometric solution of the equationax−x2 =b2. Ofterdinger38 (p. 15) seems to have quite missed the meaning of this problem. He thought, apparently, that it was equivalent tox. 16, lemma, of the Elements. 106 R. Simson, Elements of Euclid, ninth ed., Edinburgh, 1793, pp. 335–6.

39–40] PROPOSITION 19 51 Wherefore the rectangleAH equals the given square uponb (i. e. the rect- angle AB□ AC) and has been applied to the given straight lineAB, deficient by the square HB107. Proposition 19. 40. “To divide a given triangle into two equal parts by a line which passes through a point situated in the interior of the triangle.”[Leonardo 3, p. 115, ll. 7–10.] “Let the given triangle beABC, and the given point in the interior of this triangle, D. A B C DE H K T Z It is required to draw throughD a straight line which divides the tri- angle ABC into two equal parts. Draw from the pointD a line parallel to the lineBC, as DE, and Apply to DE a rectangle equal to half of the rectangleAB□ BC, such as TB□ DE) TB = AB□ BC 2DE] . Apply to the lineTB a parallelogram equal to the rectangleBT□ BE and deficient by a square1071. [Prop. 18] 107 It is not in the manner of Euclid to take account of the two solutions found by considering (F), as well asZ, determined by the circle with centreO. Although Leonardo’s construction for Problem 19 is identical with that of Euclid who makes use of Problem 18, Leonardo does not seem to have anywhere formulated Problem 18. He may have considered it sufficiently obvious fromvi. 28, or fromii. 5 andii. 6, of which he gives the enunciations in the early pages (15–16) of hisPractica Geometriae; he also considers (p. 60) the roots of a resulting quadratic equation,ax−x2 = b (cf. Cantor,Vorlesungen... ,ii2, 39), butdoesnotgive ii.14. Cf. Bibliotheca Mathematica, (3), 1907–8,viii, 190; and alsoix, 245.

52 EUCLID’S BOOK ON DIVISION OF FIGURES III [40 Let the rectangle applied be BH□ HT [(TB− HT)□ HT = TB□ BE]. Draw the lineHD and produce it toZ. Then this is the line required and the triangleABC is divided into two equal parts HBZ and HZCA. Demonstration. Therectangle TB□BE isequaltotherectangle TH□HB, whence it follows that BT : T H = HB : BE ; then dividendo108 T B : BH = BH : HE. But BH : HE = BZ : ED; [vi. 2] therefore T B : BH = BZ : ED. Consequently the rectangleTB□ED is equal to the rectangleBH□BZ. But the rectangle TB□ ED is equal to half the rectangleAB□ BC; and BH□ BZ : AB□ BC =△HBZ :△ABC 88, since the angleB is common. The triangleHBZ is, then, half the triangle ABC. Therefore the triangle ABC is divided into two equal partsBHZ and AHZC. If, in applying toTB a parallelogram equal to the rectangleTB□BE and of which the complement is a square, we obtain the rectangleAB□ AT109, 1071 The corresponding sentence in Leonardo is (p. 115, ll. 15–17): “Deinde lineegz applicabis paralilogramum deficiens figura tetragona, quod sit equale superficiesge in gz.” 108 “Elements, Bookv, definition 16” (Woepcke). This is definition 15 inHeath, The Thirteen Books of Euclid’s Elements, ii, 135. 109 “In other words whenH coincides withA. This can only be the case whenD is situated on the line which joinsA to the middle of the baseBC” (Woepcke). IfD were at the centre

of gravity of the triangle, three lines could be drawn throughD dividing the triangle into two equal parts. As introductory to his Prop. 3, Leonardo proved that the medians of a triangle meet in a point, and trisect one another—results known to Archimedes601, but no complete, strictly geometric proof has come down to us from the Greeks. Leonardo then proves that if a point be taken on any one of the medians, or on one of the medians produced, the line through this point and the corresponding angular point of the triangle will divide the triangle into two equal parts. He next shows that lines through the vertices of a triangle and any point within not on one of the medians, will divide the given triangle into triangles whose areas are each either greater than or less than the area of half of the original triangle. This leads Leonardo to the consideration of the problem, to draw through a point, within a triangle and not on one of the medians, a line whichwill bisect the area of the triangle. (Euclid, Prop. 19.) The last paragraph of Euclid's proof, as it has come down to us through Arabian sources, does not ring true, and it was not in the Euclidean manner to consider special cases. After Leonardo's proof of Proposition 19, a numerical example is given.

40–41] PROPOSITION 20 53 we may demonstrate in an analogous manner, by drawing the lineAD and prolonging it toK, that the triangleABK is one-half of the triangleABC. And this is what was required to be demonstrated." Proposition 20. 41. "To cut off a certain fraction from a given triangle by a line drawn from a given point situated in the interior of the triangle."[Leonardo 10, p. 121, ll. 1–2.] "Let ABC be the given triangle andD the given point in the interior of the triangle. It is required to pass through the pointD a straight line which cuts off a certain fraction of the triangleABC. A B C DE H T Z "Let the certain fraction be one-third. Draw from the pointD a line parallel to the line BC, as DE, and apply to DE a rectangle equal to one-third of the rectangleAB□ BC. Let this be BZ□ ED) BZ = AB□ BC 3□ ED] . Then apply toZB a rectangle equal to the rectangleZB□ BE and deficient by a square. [Prop. 18.] Let the rectangle applied be the rectangle BH□ HZ [(ZB− HZ) HZ = ZB□ BE] . Draw the lineHD and produce it toT. "On proceeding as above we may demonstrate that the triangleHTB is one-third of the triangleABC; and by means of an analogous construction to this we may divide the triangle in any ratio. But this is what it is required to do110 ." 110 Leonardo gives the details of the proof for the case of one-third and does not refer

54 EUCLID'S BOOK ON DIVISION OF FIGURES III [41–42 Proposition 21. 42. "Given the four linesA, B, C, D and that the product ofA and D is greater than the product ofB and C; I say that the ratio ofA to B will be greater than the ratio ofC to D" 111. to any other fraction. If, however, the "certain fraction" were the ratio of the lengths of two given lines,m :n, we could readily construct a rectangle equal tom n□AB□BC, and then find the rectangleBZ□ ED equal to it. The rest of the construction is the same as given above. According to the conditions set forth in Proposition 18, there will be two, one, or no solutions of Propositions 19 and 20. Leonardo considers only the Euclidean cases.Cf. notes 104 and 107. The case where there is no solution may be readily indicated. Suppose, in the above figure, that BE = EH, then of all triangles formed by lines drawn throughD to meet AB and BC, the triangleHBT has the minimum area. (Easily shown synthetically as in D. Cresswell, An Elementary

Treatise on the Geometrical and Algebraical Investi- gations of Maxima and Minima. Second edition, Cambridge, 1817, pp. 15–17.) Similar minimum triangles may be found in connection with the pairs of sidesAB, AC and AC, CB. Suppose that neither of these triangles is less than the triangleHBT. Then if △HBT :△ABC >m :n, the solution of the problem is impossible. 111 This and the next four auxiliary propositions for which I supply possible proofs, seem to be neither formally stated nor proved by Leonardo. At least some of the results are nevertheless assumed in his discussion of Euclid's later propositions, as we shall presently see. Although these auxiliary propositions are not given in theElements, they are assumed as known by Archimedes, Ptolemy and Apollonius. For example, in Archimedes' "On Sphere and Cylinder,"ii. 9 (Heiberg, ed.i, 1910, p. 227; Heath, ed. 1897, p. 90), Woepcke 21 is used. See also Eutocius' Commentary (ArchimedisOpera omniaed.Heiberg, iii, 1881, p.257, etc.), andHeiberg,Quaestiones Archimedeae, Hauniae, 1879, p. 45 f. For a possible application by Archimedes (in his Measurement of a circle) of what is practically equivalent to Woepcke 24, see Heath's Archimedes... , 1897, p. xc. The equivalent of Woepcke 24 is assumed in the proof of a proposition given by Ptolemy (87–165 A. D.) in hisSyntaxis, vol.i, Heiberg edition, Leipzig, 1898, pp. 43– 44. This in turn is tacitly assumed by Aristarchus of Samos (circa310–230 B. C.) in his work On the Sizes and Distances of the Sun and Moon(see Heath's editionAristarchus of Samos the Ancient Copernicus, Oxford, 1913, pp. 367, 369, 377, 381, 389, 391). As to the use of the auxiliary propositions in the two worksProportional Section and On Cutting off a Space, of Apollonius, we must refer to Pappus' account (Pappi Alexandrini Collectionis... ed. Hultsch, vol. ii, 1877, pp. 684 ff.). Woepcke 21, 22 occur on pp. 696–697; Woepcke 24 enters on pp. 684–687; Woepcke 23, 25 are given on pp. 687, 689. Perhaps this last statement should be modified; for whereas Euclid's propositions affirm that if a :b□c :d, a −b :b□c−d :d, Pappus shows that if a :b□c :d, a :a−b□c :c−d;

42] PROPOSITION 21 55 GivenA□D>B □C. To proveA :B >C :D. but these propositions are immediately followed by others which state that if a :b□c :d, then b :a□d :c. Below is given a list of the various restorations of the above-named works of Apol- lonius, based on the account of Pappus. By reference to these restorations the way in which the auxiliary propositions are used or avoided may be observed. We have already (Art. 21) noticed a connection of Apollonius' workOn Cutting off a Spacewith our subject under discussion. Some of these titles will therefore supplement the list given in the Appendix. Wilebrordi Snellii R. F.περὶ λόγου ἀποτομ῀ες καὶ περὶ ςηωρίου ἀποτομης (Apollonii) resuscitata geometria.Lugodini, ex officina PlatinianaRaphelengii, md.cvii pp. 23. More or less extensive abridgment of Snellius's work is given in: (a) Universae geometriae mixtaeque mathematicae synopsis et bini refractionum demonstratarum tractatus. Studio et opera F. M. Mersenni.Parisiis,m.dc.xliv, p. 382. (b) Cursus mathematicus, P. Herigone. Paris, 1634, tomei, pp. 899–904; also Paris, 1644. Apollonii Pergaei de sectione rationis libri duo ex ArabicoMS to Latine versi ac- cedunt ejusdem de sectione spatii libri duo restituti...opera & studio Edmundi Hal- ley...Oxonii,... mdccvi, pp. 8 + liii + 168. (a) Die Bücher des Apollonius von Perga De sectione rationis nach dem Lateinischen des Edm. Halley frey bearbeitet, und mit einem Anhange versehen von W. A. Diesterweg,

Berlin, 1824, pp. xvi + 218 + 9 pl. (b) Des Apollonius von Perga zwei Bücher vom Verhältnissschnitt(de sectione ratio- nis) aus dem Lateinischen des Halley übersetzt und mit Anmerkungen begleitet und einem Anhang versehen von August Richter...Elbing, 1836, pp. xxii + 143 + 4 pl. Die Bücher des Apollonius von Perga de sectione spatii wiederhergestellt von Dr W. A. Diesterweg... Elberfeld, 1827...pp. vi + 154 + 5 pl. Des Apollonius von Perga zwei Bücher vom Raumschnitt. Ein Versuch in der alten Geometrie von A. Richter.Halberstadt, 1828, pp. xvi + 105 + 9 pl. Die Bücher des Apollonius von Perga de sectione spatii, analytisch bearbeitet und mit einem Anhange von mehreren Aufgaben ähnlicher Art versehen von M. G. Grabow... Frankfurt a. M., 1834, pp. 80 + 3 pl. Geometrische Analysis enthaltend des Apollonius von Perga sectio rationis, spatii und determinata, nebst einem Anhange zu der letzten, neu bearbeitet vom Prof. Dr Georg Paucker, Leipzig, 1837, pp. xii + 167 + 9 pl. M. Chasles discovered that by means of the theory of involution a single method of solution could be applied to the main problem of the three books of Apollonius above mentioned. This solution was first published inThe Mathematician, vol.iii, Nov. 1848, pp. 201–202. This is reproduced by A. Wiegand in hisDie schwierigeren geometrischen Aufgaben aus des Herrn Prof. C. A. Jacobi Anhängen zu Van Swinden's Elementen der Geometrie. Mit Ergänzungen englischer Mathematiker... Halle, 1849, pp. 148– 149, and it appears at greater length in Chasles'Traité de Géométrie supérieure, Paris,

56 EUCLID'S BOOK ON DIVISION OF FIGURES III [42–43 Let the linesA,D be adjacent sides of a rectangle; and let there be another rectangle with sideB lying alongA and sideC alongD. Then eitherA is greater than B, orD greater thanC, for otherwise the rectangleA□D would not be greater than the rectangleB□C. Letthen A>B . ToD applytherectangle B□C andwegetarectangle A′□D =B□C; [i. 44–45] then A′ :B =C :D. [vii. 19] But since A>A ′, A :B >A′ :B; [v. 8] □A :B >C :D. [v. 13] q. e. d. Pappus remarks: Conversely ifA : B > C: D, A□D > B□C. The proof follows at once. For, findA′ such that A′ :B =C :D; then A :B >A′ :B, and A>A ′. ButA′□D =B□C. □A□D>B □C. q. e. d. Proposition 22. 43. "And when the product ofA and D is less than the product ofB and C, then the ratio ofA to B is less than the ratio ofC to D." From the above proof we evidently have C :D>A :B, that is, A :B <C :D. Conversely, as above, ifA :B <C :D, A□B <C□D. 1852, pp. 216–218;2e éd. 1880, pp. 202–204. It was no doubt Chasles who inspiredDie Elemente der projectivischen Geometrie in synthetischer Behandlung. Vorlesungen von H. Hankel, (Leipzig, 1875), "Vierter Abschnitt, Aufgaben des Apollonius," pp. 128–145; "sectio rationis," pp. 128–138; "sectio spatii," pp. 138–140. The "Three Sections," the "Tangencies" and a "Loci Problem" of Apollonius...by M. Gardiner, Melbourne, 1860. Reprinted from theTransactions of the Royal Society of Victoria, 1860–1861,v, 19–91 + 10 pl. Die sectio rationis, sectio spatii und sectio determinata des Apollonius nebst eini- gen verwandten geometrischen Aufgaben von Fr. von Lühmann. Progr. Königsberg in d. N. 1882, pp. 16 + 1 pl. "Ueber die fünf Aufgaben des Apollonius,"von L. F. Ofterdinger. Jahreshefte des Vereines für Math. u. Naturwiss. in Ulm a. D.1888, i, 21–38; "Verhältnissschnitt," pp. 23–25; "Flächenschnitt," pp. 26–27.

43–45] PROPOSITION 23–24 57 Itisreallythisconverse, andnottheproposition, whichEuclidusesinPropo- sition 26. Proclus remarks (page 407) that the converses of

Euclid’s Elements, i. 35, 36, about parallelograms, are unnecessary “because it is easy to see that the method would be the same, and therefore the reader may properly be left to prove them for himself.” No doubt similar comment is justifiable here. Proposition 23. 44. “Given any two straight lines and on these lines the pointsA, B, and D, E; and let the ratio of AB to BC be greater than the ratio of DE : EZ; I say thatdividendo the ratio ofAC to CB will be greater than the ratio ofDZ to ZE.” AB C DE [W] Z Given AB :BC >DE :EZ. To prove AC :CB >DZ :ZE. To AB, BC, DE find a fourth proportionalEW. [vi. 12] Then AB :BC =DE :EW. .....(1) But AB :BC >DE :EZ; ∴DE :EW >DE :EZ. [v. 13] ∴EW <EZ. [v. 8] From (1) AC :CB =DW :WE ; .....(2) [v. 17] since DW >DZ, DW :WE >DZ :WE. [v. 8] ∴AC :CB >DZ :WE. [v. 13] But WE <ZE ; ∴DZ :WE >DZ :ZE. [v. 8] ∴AC :CB >DZ :ZE. From (2) and [v. 17] q. e. d. Proposition 24. 45. “And in an exactly analogous manner I say that when the ratio of AC to CB is greater than the ratio ofDZ to ZE, we shall havecompo- nendo112 the ratio ofAB to BC is greater than the ratio ofDE to EZ.” 112 “Elements, Bookv, definition 15” (Woepcke). This is definition 14 inHeath, The Thirteen Books of Euclid’s Elements, ii, 135.

58 EUCLID’S BOOK ON DIVISION OF FIGURES III [45–46 Given AC :CB >DZ :ZE. To prove AB :BC >DE :EZ. Determine W, as before, such that AB :BC =DE :EW. Then AC :CB =DW :WE. [v. 17] ∴ DW :WE >DZ :ZE. .....(1) [v. 13] Now either EW >EZ or EW <EZ. If EW >EZ, DW <DZ, and DW :EW <DZ :EW. [v. 8] So much the more is DW :EW <DZ :EZ [v. 8] which contradicts (1). ∴EW <EZ. But AB :BC =DE :EW, and DE :EW >DE :EZ; [v. 8] ∴AB :BC >DE :EZ. [v. 13] q. e. d. Proposition 25. 46. “Suppose again that the ratio ofAB to BC were less than the AB C DE Z ratio of DE to EZ; dividendo the ratio ofAC to CB will be less than the ratio of DE to ZE 113.” Just as the proof of Proposition 22 was contained in that for Proposition 21, so here, the proof required is contained in the proof of Proposition 23. Similarly the converse of Proposition 25 flows out of 24. 113 The auxiliary propositions are introduced, apparently, to assist in rendering, with faultless logic, the remarkable proof of Proposition 26. In this proof it will be observed that we are referred back to Proposition 21, to the converse of Proposition 22 and to Proposition 25 only, although 23 is really the same as 25. But no step in the reasoning has led to Proposition 24. If this is unnecessary, why has it been introduced?

46–47] PROPOSITION 26 59 Proposition 26. 47. “To divide a given triangle into two equal parts by a line drawn from a given point situated outside the triangle.” [Leonardo 4, p. 116, ll. 35–36.] Let the triangle beabg and d the point outside. Join ad and let ad meet bg in e. If be = eg, what was required is done. For the trianglesabe, aeg being on equal bases and of the same altitude are equal in area. But if be be not equal to eg, let it be greater, and draw throughd, parallel to bg, a line meetingab produced in z. Since be > 1 2bg, area ab∆ be > 1 2 area ab∆ bg; [Cf. vii. 17] much more then is area ab∆ zd > 1 2 area ab∆ bg, since zd > be. Now take area ib∆ zd = 1 2 area ab∆ bg; [i. 44] then area ab∆ be > area ib∆ zd, and zd : be < ba : bi 114. [Prop. 21 or 22] But zd : be = za : ab, [vi. 4] ∴ zb : ba < ai : ib; [v. 13 and Prop. 25] or area zb∆ bi < area ba∆ ai. [Converse of Prop. 22] To answer this question, let us inspect the auxiliary propositions more closely. In a sense Propositions 21 and 22 go together: Ifad∆ bc, thena : b∆ c : d. So also for Propositions 23 and 25: Ifa : b∆ c : d,

then a−b : b□ c−d : d. Proposition 24 is really the converse of 23: Ifa : b > c: d, then a +b : b > c+d : d. Had Euclid given another proposition: If a : b < c : d then a +b : b < c+d : d, we should have had two groups of propositions 21, 22, and 23, 25 with their converses. Now the converses of 21 and 22 are exceedingly evident in both statement and proof. But this can hardly be said of the proof of 24, the converse of 23. The converse of 23 having been given the formulation of the statement and proof of the converse of 25 is obvious and unnecessary to state, according to Euclid's ideals (cf. Art. 43). It might therefore seem that Proposition 24 is merely given to complete what is not altogether obvious, in connection with the statement of the four propositions 21 and 22, 23 and 25, and their converses. In Pappus' discussion some support is given to this view, since Propositions 21 and 22 and converses are treated as a single proposition; Propositions 23, 25 as another proposition, while the converses of 23 and 25 are dealt with separately. The more probable explanation is, however, that Propositions 23 and 24 were given by Euclid because they were necessary for the discussion of other cases of Proposition 26 (assuming that the first case of Leonardo was that given by Euclid), for it was not his manner to consider different cases. Indeed if we takebe less than ge in the first part of Leonardo's discussion exactly Propositions 23 and 24 are necessary.

60 EUCLID'S BOOK ON DIVISION OF FIGURES III [47 Apply a rectangle equal to the rectangle zb□ bi to the line bi, but exceedingbyasquare 115; thatisto bi applyalinesuchthatwhenmultiplied by itself and bybi the sum will be equal to the product ofzb and bi; let ti be the side of the square116. 114 Therefore bi < ba, and ifbi be measured alongba, i will fall betweenb and a. 115 Here again we have an expression with the true Greek ring: "adiungatur quidem recte□bi□ paralilogramum superhabundans figura tetragona equale superficiei□zb□ in□bi□" 116 We we have seen thati lies betweenb and a. And since it has been shown that zb□ bi < ba□ ai, we now haveba□ ai > bt□ ti. If bt > ba, ti is also greater thanai, and bt□ ti □ ba□ ai. Therefore bt < ba andt falls betweenb anda. But it also falls between a and i by reason of the construction (always possible) which is called for. In his book onDivisions (of figures) Euclid does not formulate the proposition here quoted, possibly because of its similarity to Proposition 18 (see note 103). A B C D Q If we let the rectanglezb□ bi =c2, ti =x, and bi =a, we have to solve geometrically the quadratic equation: ax +x2 =c2. Heath points out (Elements, vol.i, pp. 386–387) that the solution of a problem the- oretically equivalent to the solution of a quadratic equation of this kind is presupposed in the fragment of Hippocrates'Quadrature of lunes(5th centuryb. c.) preserved in a quotation by Simplicius (fl. 500a. d.) from Eudemus'History of Geometry(4th century b. c.). See Simplicius'Comment. in Aristot. Phys.ed. H. Diels, Berlin, 1882, pp. 61–68; see also F. Rudio, Der Bericht des Simplicius über die Quadratur des Antiphon und Hippokrates, Leipzig, 1907. Moreover as Proposition 18 is suggested by theElements, ii. 5, so here this problem is suggested byii. 6: If a straight line be bisected and a straight line be added to it in a straight line, the rectangle contained by the whole with the added straight line and the added straight line together with the square on the half is equal to the square on the straight line made up of the half and the added straight line.

47] PROPOSITION 26 61 Draw the straight linetkd. Since a b d e g i k t z area zb□ bi = bi□ ti + ti 2 = area bt□ ti, zb : bt = ti : ib, [vii. 19] or zt : bt = bt : bi. [v. 18] But zt : bt = zd : bk, [vi. 4] □ zd : bk = bt : bi, and area kb□ bt = area zd□ bi. But area zd□ bi = 1 2 area ab□ bg, □△tbk = 1 2△abg88. If AB is the straight line bisected atC and BD is the straight line added, then by ii. 6, AD□DB +CB 2 =CD 2. In his solution of our problem, Robert Simson proceeds, in effect, as follows (Elements of Euclid, ninth ed., Edinburgh, 1793, p. 336): DrawBQ at right angles toAB and equal toc. Join CQ and describe a circle with centreC and radius CQ cutting AB produced in D. Then BD or x is found. For, byii. 6, AD□DB +CB 2 =CD 2, =CQ 2, =CB 2 +BQ2, □AD□DB =BQ2, whence (a +x)x =c2 or ax +x2 =c2. It was not Euclid's manner to consider more than one solution in this case.

62 EUCLID'S BOOK ON DIVISION OF FIGURES III [47 Therefore the triangleabg is divided by a line drawn from the pointd, that is, by the linetkd, into two equal parts one of which is the triangle tbk, and the other the quadrilateraltkga. q. e. f. Leonardo now gives a numerical example. He then continues: [If the pointd were on one side,ab, produced at say,z], throughz draw ze parallel to bg and meeting ag produced in e. a b e g i l t z Make area ze□ gi = 1 2 area ag□ gb, and apply a rectangle, equal to the rectangleeg□ gi, to the line gi, but exceeded by a square; then eg□ gi = gt□ ti. Join tz, then [this is the required line. The proof is step for step as in the first case]. Leonardo then remarks: "Que etiam demonstrentur in numeris," and pro- ceeds to a numerical example. Thereafter he continues: a b c d e g i k l t z But let the sidesab, gb of the triangle be produced tod and e respec- tively; and let i be the given point in the angleebd from which a line is to be drawn dividing the triangleabg into two equal parts. Join ib and

47–48] PROPOSITION 27 63 produce it to meetag in z. If az = zg, the triangleabg is divided into two equal parts by the lineiz. But [if az > zg,] let za produced meet, in the point t, the line drawn throughi parallel to ab. Since za > 1 2ag, area ab□ az > 1 2 area ba□ ag. Make area it□ ak = 1 2area ba□ ag; then make area al□ kl = area ta□ ak. Join il. Then as above the triangleabg is divided into two equal parts by the lineil, one part the trianglelac, the other the quadrilaterallcbg. To this statement Leonardo adds nothing further. The proof that k lies betweena and z, andl betweenk and z, follows as in the first part. Proposition 27. 48. "To cut off a certain fraction of a triangle by a straight line drawn from a given point situated outside of the triangle117 ." [Leonardo11, p.121, ll. 22–23.] Let abgbethegiventriangleand dthegivenpointoutside. Itisrequired to cut off from the triangle a certain fraction, say one-third, by a line drawn through d. Join ad, cutting bg in c. If either bc or cg be one-third of bg, then the linead through the pointd cuts off one-third of the triangleabg. But if this be not the case produceab, ag to meet inz and e respectively the line drawn throughd parallel to bg. 117 Some generalizations of the triangle problems in Propositions 19, 20, 26 and 27 may be remarked. Steiner, in 1827, solved the problem: through a given point on a sphere to draw an arc of a great circle cutting two given great circles such that the intercepted area is equal to a given area.(J. Steiner, "Verwandlung und Theilung sphärischen Figuren durch Construction,"Crelle Jl, ii (1827), pp. 56 f. Cf. Syllabus of Townsend's course at Dublin Univ., 1846, inNouvelles Annales de Mathématiques, Sept. 1850,ix, 364; also Question 427(7) proposed by Vannson

inNouvelles Annales... Jan. 1858,xvii, 45; answered Aug. 1859,xviii, 335–6.) See alsoGudermann, “Über die niedere Sphärik,”Crelle Jl, 1832,viii, 368. In the next year Bobillier solved, by means of planes and spheres only, the problem, to draw through a given line a plane which shall cut off from a given cone of revolution a required volume. (Correspondance Math. et Physique(Quetelet), vie livraison, iv, 2–3, Bruxelles, 1828.)

64 EUCLID’S BOOK ON DIVISION OF FIGURES III [48 a b c d e g i k m z Make area de□ gi = 1 3 area ag□ gb, and apply to the line gi a rectangle equal to the rectangle eg□ gi, but exceeded by a square; then eg□ gi = ik□ kg. Draw the linekmd. I say that the trianglekmg is one-third of the triangle abg. Proof: For since area eg□ gi = area gk□ ki, eg : gk = ki : ig. Hence ek : gk = gk : gi. [v. 17] But ek : kg = de : gm; [vi. 2] □ ed : gm = gk : gi. □ area gk□ gm = area de□ gi. But area de□ gi = 1 3 area ag□ gb; □ area gk□ gm = 1 3 area ag□ gb. And since area gk□ gm : area ag□ gb =△kgm :△agb88, △kgm = 1 3△agb. In a similar manner any part of a triangle may be cut off by a straight line drawn from a given point, on a side of the triangle produced, or within two produced sides.

48–49] PROPOSITION 28 65 Proposition 28. 49. “To divide into two equal parts a given figure bounded by an arc of a circle and by two straight lines which form a given angle.”[Leonardo 57, p. 148, ll. 13–14.] “Let ABC be the given figure bounded by the arcBC and by the two lines AB, AC which form the angleBAC. It is required to draw a straight line which will divide the figureABC into two equal parts. A BC E T Z “Draw the line BC and bisect it atE. Through the point E draw a line perpendicular toBC, asEZ, and draw the lineAE. Then becauseBE is equal toEC, the areaBZE is equal to the areaEZC, and the triangle ABE is equal to the triangleAEC. Then the figureABZE will equal the figure ZCAE. If the lineAE lie inEZ produced, the figure will be divided into two equal partsABZE and CAEZ. But if the lineAE be not in the line ZE produced, join A to Z by a straight line and through the point E draw a line, asET, parallel to the lineAZ. Finally draw the lineTZ. I say, that the lineTZ is that which it is required to find, and that the figure ABC is divided into two equal partsABZT and ZCT. “For since the two trianglesTZA and EZA are constructed on the same base AZ and contained between the same parallelsAZ, TE: the triangle ZTAis equal to the triangleAEZ. Then, adding to each the common part AZB, we have TZBA equal to ABZE. But this latter figure was half of the figure ABC; consequently the lineZT is the line sought andBZCA is divided into two equal partsABZT , TZC, which was to be demonstrated.” Leonardo’s proof is practically word for word as the above. He gives two figures and in each he uses theGreek succession of letters. It is doubtless to this Proposition and the next that reference is made in the account of Proclus [Art.i].

66 EUCLID’S BOOK ON DIVISION OF FIGURES III [49–50 Proposition 29. 50. “To draw in a given circle two parallel lines cutting off a certain fraction from the circle.” [Leonardo 51 (the case where the fraction is one-third), p. 146, ll. 37–38.] “Let the certain fraction be one-third, and the circle beABC. It is required to do that which is about to be explained. A B C E D Z “Construct the side of the triangle (regular) inscribed in this circle. Let this be AC. Draw the two linesAD, DC and draw through the pointD a line parallel to the lineAC, such asDB. Draw the lineCB. Divide the arc

AC into two equal parts at the pointE, and draw from the pointE the chord EZ parallel to the lineBC. Finally draw the lineAB. I say that we have two parallel linesEZ , CB cutting off a third of the circleABC, viz. the figure ZBCE. “Demonstration. The lineAC being parallel to the lineDB, the triangle DAC will be equal to the triangleBAC; add to each the segment of the circle AEC; the whole figure DAEC will be equal to the whole figure BAEC. But the figureDAEC is one-third of the circle. Consequently the figure BAEC is also one-third of the circle. SinceEZ is parallel to CB, the arc EC will be equal to the arcBZ; but EC is equal toEA, hence EA equals ZB. Add to these equal parts the arcECB; the whole arcAB will equal the whole arc EZ. Consequently the line AB will be equal to the line EZ, and the segment of the circleAECB will be equal to the segment of the circleECBZ. Taking away the common segmentBC, there remains the figure EZBC equal to the figure BAEC. But the figure BAEC was one-third of the circle ABC. Then the figure EZBC is one-third of the circle ABC; which was to be demonstrated.

50] PROPOSITION 29 67 “When it is required to cut off a quarter of a circle, or a fifth or any other definite fraction, by means of two parallel lines, we construct in this circle the side of a square or of the pentagon (regular) inscribed in the circle and we draw from the centre to the extremities of this side the two straight lines as above. (The remainder of) the construction will be analogous to that which has gone before118 The statement and form of discussion of this proposition are not wholly satisfactory. For “a certain fraction” in the enunciation we should rather expect “one-third,” as in Leonardo; while at the conclusion of the proof might possibly occur a remark to the effect that a similar construction would apply when the certain fraction was one-quarter [by means ofiv. 6], one-fifth [iv. 11], one-sixth [iv. 15], or one-fifteenth [iv. 16], but is it conceivable that Euclid added “or any other definite fraction”? Moreover the lack of definition ofD and certain matters of form seem to further indicate that modification of the original has taken place in its passage through Arabian channels. On the other hand Leonardo presents the proposition as if drawn from the pure well of Euclid undefiled. Here is his discussion. (I have substitutedC for his b, andB for hisg.) “And if, by means of two parallel lines, we wish to cut off from a circleACB, whose centre isD, a given part which is one-third, draw the lineAC, the side of an equilateral triangle inscribed in the circleabg. Through the centreD draw DB parallel to this line and joinCB. Bisect the arc AC at E and draw EZ parallel tobg. I say that the figure contained between the linesCB and EZ and the arcs EC and BZ is one-third part of the circleACB. “Proof: Draw the linesDA and DB and AB. “The triangles BAC and DAC are equal. To each add the portion ABE. Then the figure bounded by the linesBA and BC and the arcAEC is equal to the sector DAEC which is a third part of the circleABC. “Therefore the figure bounded by the linesBA and BC and the arcAEC is a third part of the circle. “And since the linesCB and EZ are parallel, the arcsEC and BZ are equal. But arc EC is equal to arcAE. Therefore arc AE is equal to the arcBZ. To each add the arcEB, and then the arcAECB will be equal to the arcECBZ. “Hence the portionEZBC of the circle is equal to the portionABCE. Take away the common part between the lineCB and the arcBC and there remains the figure, bounded by the linesBC and EZ and the arcsCE and BZ, which is 118 This problem is clearly not susceptible of solution with ruler and compasses, in such a

case as when the “certain fraction,”1 n, is one-seventh. In fact the only cases in which the problem is possible, for a fraction of this kind, is whenn is of the form 2p(22s1 + 1)(22s2 + 1)... (22sm + 1), wherep, ands’s (all different), are positive integers or zero, and22sm +1(m = 1, 2,...m) is a prime number. (Cf. C. F. Gauss,Disquisitiones Arithmeticae, Lipsiae, 1801, French ed., Paris, 1807, p. 489.)

68 EUCLID’S BOOK ON DIVISION OF FIGURES III [50–51 the third part of the circle since it is equal to the figure bounded by the lines BA and BC and the arcAEC; quod oportebat ostendere.” In hisμετρικά (iii. 18) Heron of Alexandria considers the problem:To divide the area of a circle into three equal parts by two straight lines. He remarks that “it is clear that the problem is not rational”; nevertheless “on account of its practical use” he proceeds to give an approximate solution. By discussion similar to that above he finds the figureBCEA, formed by the triangleBCA and the segment CEA, to be one-third of the circle. Neglecting the smaller segment with chord BC, we have, thatBA cuts off “approximately” one-third of the circle. Similarly a second chord fromB might be drawn to cut off another third of the circle, and the approximate solution be completed. Proposition 30. 51. ”To divide a given triangle into two parts by a line parallel to its base, such that the ratio of one of the two parts to the other is equal to a given ratio.” a b e g h i k z Although Leonardo does not explicitly formulate this problem or the next, the method to be employed is clearly indicated in the discussion of Proposition 5 (Art. 26). Let abg be the triangle which is to be divided in the given ratioez : zi, by a line parallel tobg. Divide ab in h such that ah2 : ab2 = ez : ei 92. Draw hk ‖ bg and meeting ag in k. Then the triangles ahk and abg are similar and △ahk :△abg = ah2 : ab2. [vi. 19] But ah2 :ab2 = ez : ei, □△ahk :△abg = ez : ei; whence △ahk : quadl.hbgk =ez :zi; [v. 16, 17] and the triangleabg has been divided as required.

51–53] PROPOSITION 31–32 69 Proposition 31. 52. “To divide a given triangle by lines parallel to its base into parts which have given ratios to one another.” a b e g h i k l m t z Again in the manner of Proposition 5, suppose it be required to divide the triangle abg into three parts in the ratio ez : zt : ti. Then determine the pointsh, l in ab such that ah2 :ab2 =ez :ei92; and al2 :ab2 =et :ei. Then ah2 :al2 =ez :et. [v. 16, 20] □△ahk :△alm = ez : et, and □△ahk : quadl.hlmk = ez : zt. Similarly △alm : quadl.lbgm = et : ti. But △ahk :△alm = ez : et. □△ahk : quadl.lbgm = ez : ti. [v. 20] Hence, △ahk : quadl.hlmk : quadl.lbgm = ez : zt : ti, and the triangle abg has been divided into three parts in a given ratio to one another. So also for any number of parts which have given ratios to one another. Proposition 32. 53. “To divide a given trapezium by a line parallel to its base, into two parts such that the ratio of one of these parts to the other is equal to a given ratio.”[Leonardo 29, p. 131, ll. 41–42.] Let abgd be the trapezium which is to be divided in the ratioez : zi by a line parallel to the base. Produce the sidesba, gd through a and d to meet in t.

70 EUCLID’S BOOK ON DIVISION OF FIGURES III [53 t a b d e g h i k l m z Make tl2 : at2 = zi : ez92, and ht2 : (bt2 + tl2) = ez : ei. Through l, h, draw lm, hk parallel to bg and ad. Then I say that the quadrilateral ag is divided in the given ratio,ez : zi, by the linehk. Proof: For since the trianglestlm, tad are similar tl2 : at2 =△tlm :△tad; but tl2 : at2 = zi : ez; □ zi : ez =△tlm :△tad. Whence ei : ez = (△tlm +△tad) :△tad, [v. 18]

or ez : ei =△tad : (△tlm +△tad). [v. 16] But by construction ez : ei = ht 2 : (bt 2 + tl 2), and ht2 : (bt2 + tl2) =△thk : (△tbg +△tlm). [vi. 19] □ ez : ei =△thk : (△tbg +△tlm). But △thk =△tad +quadl.ak. Similarly △tbg +△tlm = quadl.ag +△tad +△tlm. □ ez : ei = (quadl.ak +△tad) : (quadl.ag +△tad +△tlm).

53] PROPOSITION 32 71 But ez : ei =△tad : (△tad +△tlm); □ ez : ei = quadl.ak : quadl.ag; [v. 11, 19] whence ez : zi = quadl.ak : quadl.hg. And the trapezium has been divided in the given ratio. Then follows a numerical example and this alternative construction and proof: Draw mls such that, ms : ls = tb2 : ta2, 92 and divide ml in n, such thatln is to nm in the given proportion. a b d g h kl m n s t In tb determine h such that th2 : tb2 = ns : sm. Draw hk‖ bg. Then, quadl.ak : quadl.hg = ln : nm.

72 EUCLID'S BOOK ON DIVISION OF FIGURES III [53–54 Proof: For, tb2 : ta2 =△tbg :△tad; [vi. 19] and ms : ls = tb2 : ta2. □ ms : ls =△tbg :△tad. . . . . [1] Again, since tb2 : th2 = ms : sn, and △tbg :△thk = tb2 : th2, □ ms : ns =△tbg :△thk; . . . . . .[2] □ sm : nm =△tbg : quadl.hg. [v. 16, 21] .....[3] But sm : ls =△tbg :△tad 1181, or ms :△tbg = ls :△tad, while ms :△tbg = ns :△thk; [from [2]] □ ls : ns =△tad :△thk. . . . . [4] From [3] ms :△tbg = nm : quadl.hg. But from [4] sl : ln =△tad : quadl.ak, [v. 16, 21] □ sl :△tad = ln : quadl.ak. But from [1] ms :△tbg = sl :△tad, □ ms :△tbg = ln : quadl.ak; □ mn : quadl.hg = ln : quadl.ak; □ ln : nm = quadl.ak : quadl.hg. Hence the quadrilateralag is divided by the linehk, parallel to the base bg, in the given proportion as the numberln is to the numbernm. Which was to be done. Then follows a numerical example. Proposition 33. 54. "To divide a given trapezium, by lines parallel to its base, into parts which have given ratios to one another."[Leonardo 35, p. 137, ll. 6– 7.] Let abgd be the given trapezium and [letez : zi : it denote 1181 Such mixed ratios as these (ratios of lines to areas), and others of like kind which follow in this proof, are very un-Greek in their formation. This is sufficient to stamp the second proof as of origin other than Greek. The first proof, on the other hand, is distinctly Euclidean in character.

54] PROPOSITION 33 73 a b d e f g hi kl m n o pr st z the ratios of the three parts into which the trapezium is to be divided by lines parallel to the basebg]. Let ba, gd produced meet ink and find l such that bk2 : ak2 = tl : el. Then determine m and n such that bk 2 : km2 = tl : lz, and bk 2 : kn2 = tl : il. Through m, n draw lines mo, np parallel to bg. In the same manner as above quadl.ao : quadl.mp = ez : zi; and quadl. mp : quadl.ng = zi : it. Then follows a numerical example in which the linekfhrs, perpendicular to bg, is introduced into the figure. Here is a proof of the Proposition:

74 EUCLID'S BOOK ON DIVISION OF FIGURES III [54–55 By construction,v. 16 andvi. 19, △kbg :△kad = tl : el. ..... [1] So also △kbg :△kmo = tl : lz, ... [2] and △kbg :△knp = tl : il ..... [3] From [1] △kad :△kbg = el : tl. But from [2] △kbg :△kmo = tl : lz; hence, by [v. 20], △kad :△kmo = el : lz, ... [4] or alternately △kmo :△kad = lz : el. Hence, separando, quadl. ao :△kad = ez : el. ..... [5] So also from [2] and [3] △kmo :△knp = lz : il; and △kmo : quadl.mp = lz : iz. But from [4] △kad :△kmo = el : lz; therefore, by [v. 20], △kad : quadl.mp = el : iz ..... [6] Hence from [5], by [v. 20], quadl.ao : quadl.mp = ez : zi. Again, from [3], quadl. ng :△kbg = ti : tl; and since from [1], △kbg :△kad = tl : el, we have quadl. ng :△kad = it : el. Hence from [6], by [v. 20], we get quadl.ng : quadl.mp = it : zi, or

alternately quadl. mp : quadl.ng = zi : it. And since quadl.ao : quadl.mp = ez : zi, the trapeziumag has been divided by lines parallel to the baseag, into three parts which are in the required ratios to one another. q. e. f. Proposition 34. 55. “To divide a given quadrilateral, by a line drawn from a given vertex of the quadrilateral, into two parts such that the ratio of one of these parts to the other is equal to a given ratio.”[Leonardo 40, p. 140, ll. 36–37.] Let abcd be the given quadrilateral, andez : zi the given ratio. It is required to draw from the angled a line to divide the quadrilateral in the ratio ez : zi.

55] PROPOSITION 34 75 a b c d e i k t z Draw the diagonalac and on it findt such that ct : at = ez : zi. Draw the diagonal bd. Then if bd pass through t the quadrilateral is divided as required, in the ratioez : zi. For, △dct :△dta = ct : ta, =△cbt :△tba. ∴ ct : ta =△dcb :△abd. [v. 18] But ct : ta = ez : zi; ∴ ez : zi =△bdc :△bda; and the quadrilateralac is divided, by a line drawn from a given angle, in a given ratio. But ifbd do not pass throught, it will cutca either betweenc and t or between t and a. Consider first whenbd cuts ct. Join bt and td. Then, quadl.tbcd : quadl.tbad = ct : ta = ez : zi. Draw tk∥ bd and join dk. Then quadl.kbcd = quadl.tbcd; ∴ quadl.kbcd :△dak = ez : zi, and the linedk has been drawn as required. If the diagonalbd cut at, through t draw tl parallel to the diagonalbd. Join dl. Then as before, ct : ta = ez : zi =△dcl : quadl.abld.

76 EUCLID’S BOOK ON DIVISION OF FIGURES III [55–57 a b c d l t Hence in every case the quadrilateral has been divided as required by a line drawn fromd. Similarly for any other vertex of the quadrilateral. Proposition 35. 56. “To divide a given quadrilateral by lines drawn from a given vertex of the quadrilateral into parts which are in given ratios to one another.” Although Leonardo does not explicitly formulate this problem, the method he would have followed is clear from his discussion of the last Proposition. Let abcd be the quadrilateral to be divided, by lines drawn fromd, into three parts in the ratios to one another ofez : zi : it. e i t z Divide ca at points r, t so that cr : rt : ta = ez : zi : it. Through r, t draw lines parallel tobd, and meetingbc (or ab) inl and ab (or bc) inm. Then as abovedl, dm divide the quadrilateral as required. We may proceed in a similar manner to divide the quadrilateralabcd, by lines drawn from the angular pointd, into any number of parts in given ratios to one another. Proposition 36. 57. “Having resolved those problems which have gone before, we are in a position to divide a given quadrilateral in a given ratio or in given ratios by a line or by lines drawn from a given point situated on one of the sides of the quadrilateral, due regard being paid to the conditions mentioned above.”

57] PROPOSITION 36 77 This problem, also, is not formulated by Leonardo; but from his discussion of Euclid’s Propositions 16, 17 and of his own 41, the methods of construction which Euclid might have employed are clearly somewhat as follows. Let abcd be the given quadrilateral andg the given point. a b c dg hl m (1) Let it be required to divideabcd into two parts in the ratioez : zi by a line drawn through a pointg in the sidead. Draw dl such that△cld : quadl.lbad = ez : zi. [Prop. 34] Join gl. If gl∥ dc, join gc, then this line divides the quadrilateral as required. If gl be not parallel todc draw dh∥ gl, and meetingbc in h. Join gh. Then gh divides the quadrilateral as required. If dh fell outside the quadrilateral drawll′∥ cd (not indicated in the figure) to

meet ad in l′. Draw l′z′‖ gc to meet dc in z′. Then gz′ is the line required. The above reasoning is on the assumption thatdl meets bc in l. Suppose now it meetab in l. Join bd and draw bk such that quadl.bcdk :△kab = ez : zi. a b c d g k l m

78 EUCLID'S BOOK ON DIVISION OF FIGURES III [57 If k do not coincide withg there are two cases to consider according ask is betweeng and d or betweeng and a. Consider the former case. Throughk draw km parallel togb and meetingbc inm. Join gm. Then this is the required line dividing the quadrilateralac in such a way that quadl.abmg : quadl.gmcd = ez : zi. Similarly ifk were betweeng and a. (2) Let it be required to divide abcd into, say, three parts in the ratios ez : zi : it, by lines through any pointg in the sidead (first figure). Draw dl, dm dividing the quadrilateralac into three parts such that △amd : quadl.dmbl :△dlc = ez : zi : it. There are various cases to consider according asl andm are both onbc, both on ab, or one onab and one onbc. The method will be obvious from working out one case, say the last. Join gc, gl. If gl be parallel tocd, gc cuts off the trianglegdc such that △gdc : quadl.abcg = it : ei (= ez + zi). [v. 24] If gl be not parallel todc, draw dh parallel to gl and meeting bc in h; then gh divides the quadrilateral in such a way that quadl.gdch : quadl.ghba = it : ei. Then apply Proposition 34 to draw fromg a line to divide the quadrilateral abhg in the ratio ofez : zi. Hence fromg are drawn two lines which divide the quadrilateralabcd into three parts whose areas are in the ratios ofez : zi : it. The case whendh meets bc produced may be considered as above. We could proceed in a similar manner if the quadrilateralabcd were to be divided by lines drawn fromg, into a greater number of parts in given ratios. The enunciation of this proposition is a manifest corruption of what Euclid may have given. Such clauses as those at the beginning and end he would only have included in the discussion of the construction and proof. After the enunciation of Proposition 36, Woepcke's translation of the Arabian MS. concludes as follows: "End of the treatise. We have confined ourselves to giving the enuncia- tions without the demonstrations, because the demonstrations are easy."

IV. APPENDIX In the earlier pages I have referred to works on Divisions of Figures written before 1500. Several of these were not published till later; for example, that of Muhammed Bagdedinus in 1570, of Leonardo Pisano in 1862 and the second edition of Luca Paciuolo's "Summa" in 1523119. It has been remarked that Fra Luca's treatment of the subject was based entirely upon that of Leonardo. But, on account of priority in publication, to Paciuolo undoubtedly belongs the credit of popularizing the problems on Divisions of Figures. While few publications treat of the subject in the early part, their number increases in the latter part, of the sixteenth century. In succeeding centuries the tale of titles is enormous and no useful purpose would be served by the publication here of an even approximately complete list. It would seem, however, as if the subject matter were of sufficient interest to warrant, as completion of the history of the problems, a selection of such references in this period, (1) to standardorpopularworks, (2)tothewritingsofeminentscientistslikeTartaglia, Huygens, Newton, Kepler and Euler; (3) to special articles, pamphlets or books which treat parts of the subject; (4) to discussions of division problems requiring other than Euclidean methods for their solution. No account is taken of the extensive literature on the division of the circum- ference of a circle, from which corresponding divisions of its area readily flow. Considerations along this line may be found in:P. Bachmann,

Die Lehre von der Kreisteilung und ihre Beziehungen zur Zahlentheorie, Leipzig, 1872, 12 + 300 pp.; and inA. Mitzscherling, Das Problem der Kreisteilung, Leipzig und Berlin, 1913, 6 + 214 pp. Except for about a dozen titles, all the books or papers mentioned have been personally examined. In many cases it will be found that only a single problem (often Euclid's Propositions 19, 20, 26 or 27) is treated in the place referred to. Some titles in note 111 may also be regarded as forming a supplement to this list. 1539—W. Schmid. Das Erst [Zweit, Dritt und Viert]Buch der Geometria. Nürnberg. "Dritter Theil, von mancherley Art der Flächen, wie dieselben gemacht und ausgetheilt werden, auch wie eine Fläche in die andern für sich selbst, oder gegen einer andern in vorgenommener Proporz, geschätzt, verändert mag werden. Theilungen und Zeichnungen von Winkeln, Figuren, ordentlichen Vielecken, die letzten, wie man leicht denken kann, nicht alle geometrisch richtig. Verwandlung von Figuren." 119 A synopsis of the portion of the work on divisions of figures is given on pages 106 and 275–284 ofScritti inediti del P. D. Pietro Cossali...pubblicati da Baldassarre Boncompagni, Roma, 1857.Cf. note 61.

80 EUCLID'S BOOK ON DIVISION OF FIGURES IV 1547—L. Ferrari. A "Cartello" which begins: "Messer N. Tartaglia, già otto giorni, cioè alli 16 di Maggio, in risposta della mia replica io riceuetti la uostra tartagliata, etc." [Milan.] Dated June 1, 1547; a challenge to a mathematical disputation from L. F. to N. Tartaglia. 1547—N. Tartaglia. Terza Risposta data da N. Tartalea...al eccellente M. H. Cardano...et al eccellente Messer L. Ferraro...con la resolutione, ouer risposta de 31 quesiti, ouer quistioni da quelli a lui proposti.[Venice, 1547.] Dated July 9, 1547. For the discussion between Ferrari (1522–1565) and Tartaglia (1500–1557) 6 "cartelli" by Ferrari and 6 "Risposte" by Tartaglia were published at Milan, Venice and Brescia in 1547–48120. They contained the prob- lems and their solutions. These publications are of excessive rarity. Only about a dozen copies (which are in the British Museum and Italian Libraries) are known to exist. But they have been reprinted in:I sei cartelli di matematica disfida pri- mamente intorno alla generale risoluzione delle equazioni cubiche, di Ludovico Ferrari, coi sei contro-cartelli in risposta di Nicolo Tartaglia...comprendenti le soluzioni de' quesiti dall' una e dall' altra parte proposti...Raccolti, autografati e pubblicati da Enrico Giordani....Milano 1876. On pages 6–7 of theiii◦ cartello (Giordani's edition pp. 94–95), Questions 5 and 14, proposed by Ferrari, are:— "5. To bisect, by a straight line, an equilateral, but not equiangular, heptagon." "14. Through a point without a triangle to draw a line which will cut off a third." On pages 12 and 20 of the iii◦ Risposta Tartaglia gives the solutions and assigns due credit to the treatment of problems on the Division of Figures by Luca Paciuolo. The general subject was treated much more at length by Tartaglia in a part of his "General trattato" published in 1560. 1560—N. Tartaglia.La quinta parte del general trattato de' numeri et mis- ure. Venetia. On folio 6recto we have a section entitled "Del modo di saper dividere una figura cioè pigliar, over formar una parte di quella in forma propria." The division of figures is treated on folios 23verso–44 recto (23–32, triangles; 32–34, paral- lelograms; 34–44, quadrilateral, pentagon, hexagon, heptagon, circle without the Euclid-Proclus case). Cf. the synopsis in Scritti inediti del P. D. Pietro Cossali chierico regolare teatino pubblicati da Baldassarre Boncompagni, Roma, 1857, pp. 299–300. 1574—J.

Gutman.Feldmessung gewiss, richtig und kurz gestellt.Heidelberg. 1574—E. Reinhold. Gründlicher und wahrer Bericht vom Feldmessen, samt allem, was dem anhängig, darinn alle die Irrthum, so bis daher im Messen 120 Cf.Cantor,Vorlesungen über Gesch. d. Math.Bdii, 2teAufl., 1900, pp.490–491, where the exact dates are given.

APPENDIX 1585–1615 81 fürgeloffen, entdeckt werden. Dessgleichen vom Markscheiden, kurzer und gründlicher Unterricht. Erffurdt. “Der dritte Theil von Theilung der Aecker. Theilungen aller Figuren, auch des Kreises mit Exemplen und Tafeln erläutert.” 1585—G.B.Benedetti. Diversarum speculationum mathematicarum et physi- carum liber.Taurini. Pages 304–307. 1604—C. Clavius.Geometria Practica.Romae. Pages 276–297. 1609—J. Kepler. Astronomia nova...commentariis de motibus stellæ Mar- tis...[Pragae]. “Kepler’s Problem” occurs on p. 300 of this work (Opera Kepleried. Frisch,iii, 401). It is: “To divide the area of a semicircle in a certain ratio by a straight line drawn through a given point on the diameter or on the diameter produced.” (Cf. A. G. Kästner, Geschichte der Math. iv, 256, Göttingen, 1800;M. Cantor, Vorlesungen etc., ii, 708, Leipzig, 1900). Kepler was led to this problem in his theory of the path of the planets. It has been attacked by many mathemati- cians, notably by Wallis, Hermann, Cassini, D. Gregory, T. Simpson, Clairaut, Lagrange, Bossut, and Laplace. (Cf. G. S. Klügel, Mathematisches Wörter- buch...Erste Abtheilung, Dritter Theil, Leipzig, 1808; Article, “Kepler’s Auf- gabe.” See also C. Hutton, Philosophical and Mathematical Dictionary. New edition, London, 1815, 1, 703.) 1612—Sybrandt Cardinael.Hondert geometrische questien met hare solu- tien. Amsterdam. This work is also to be found at the end ofJohan SemsendeIan DouPracti- jck des landmetens. Amsterdam, 1616. Another edition: Tractatus geometricus. Darinnen hundert schöne ...Questien [übersetzt] durch Sebastianum Curtium. Amsterdam, 1617; Questions 78, 90–93. With these problems Huygens (1629–1695) busied himself when about 17 or 18 years of age.Cf. Oeuvres complètes de Chr. Huygens, Amsterdam, XI, 24 and 29, 1908. I have elsewhere (Nieuw Archief, 1914) shown that Sybrandt Cardinael’s work was translated into English, rearranged and published as an original work by Thomas Rudd (1584?-1656): A hundred geometrical Questions with their solu- tions and demonstrations. London, M.DC.L. 1615—Ludolph van Ceulen. Fundamenta arithmetica et geometrica cum eorundem usu ...e vernaculo in Latinum translata a W. S[nellio], R. F. Lugduni Batavorum. Contains several problems on Change, and Division, of Figures.

82 EUCLID’S BOOK ON DIVISION OF FIGURES IV 1616—J. Speidell. A geometricall Extraction or a compendiovs collection of the chiefe and choyse Problemes, collected out of the best, and latest writers. Wherevnto is added about 30 Problemes of the Authors Invention, being for the most part, performed by a better and briefer way, than by any former writer. London. Another edition, 1617; second edition “corrected and enlarged,” London, 1657; “Now followeth [pp. 84–125] a compleat Instruction of the diuision of all right lined figures.... Very pleasant and full of delight in practise: Also, most prof- itable to all surveighers, or others that are desirous to make any Inclosure.” 1619—A. Anderson. Exercitationum mathematicarum Decas prima. Con- tinens, Questionum aliquot, quae nobilissimorum tum hujus tum veteris aevi, Mathematicorum ingenia exercuere, Enodationem.

Parisiis. Problems in division of a triangle, with reference to Clavius (1604).Cf. The Ladies' Diary, London, 1840, pp. 55–56. 1645—C. Huygens. Oeuvres Complètes, xi, 1908, pp. 26–27; 219–225. Solution of "Datum triangulum, ex puncto in latere dato, bifariam secare" (1645); two solutions of "Triang. ABC, sectus utcumque lineâ DE, dividendus est aliâ lineâ, FG, ita ut utraque pars DBE et ADEC bifariam dividatur" (1650– 1668). See also note under 1687—J. Bernoulli. 1657—F. van Schooten.Exercitationvm mathematicarum liber primus con- tinens propositionum arithmeticarvm et geometricarvm centuriam. Lugd. Batav. Prop. L, pp. 107–110. Dutch edition, Amsterdam, 1659. pp. 107–110. Con- cerning a Schooten MS. of 1645–6, used by Huygens and of interest in this con- nection, cf. C. Huygens,Oeuvres Complètes, tomexi, 1908, p. 13 ff. 1667—D. Schwenter.Geometriae Practicae novae et auctae Libri IV ...mit vielen nutzlichen Additionen und neuen Figuren vermehret durch G. A. Böck- lern. Nürnberg. "Von Austheilung der Figuren in gleiche und ungleiche Theil," pp. 269–279; p. 350; the problem on this last page is taken fromB. Bramer, Trigonometria planarum mechanica, Marpurg, 1617, p. 99. "Von Austheilungen der Aecker Wiesen,..." pp. 567–583. 1674—C. F. M. Deschales.Cursus seu mundus mathematicus. Lugduni. "De figurarum planarum divisione,"i, 371–381; second edition, 1690. 1676—I. Newton. Arithmetica Universalis. Cantabrigiae,MDCCVII. Prob. X, p. 126 (Prob.XX, pp. 254–255 of the 1769 edition). This problem was discussed in a lecture delivered October, 1676 (seeCorrespondence of Sir Isaac Newton and Professor Cotes...by J. Edleston, London, 1850, p. xciii).

APPENDIX 1684–1739 83 1684—T. Strode. A Discourse of Combinations, Alternations and Aliquod Parts by John Wallis. London, 1685. On pages 163–164 is printed a letter, dated Nov. 1684, from Strode to Wallis. It discusses two problems on divisions of a triangle. 1687—J. Bernoulli."Solutio algebraica problematis de quadrisectione trian- guli scaleni, per duas normales rectas."Acta Eruditorum, 1687, pp. 617– 623. Also inOpera, Genevae, 1744, I, 328–335; see further II, 671. In the solution of this question Bernoulli is led to the intersection of a conic and a curve of the fourth degree, that is, to an equation of the eighth degree. And yet, in the seventh edition ofRouché et Comberousse, Traité de Géométrie, Paris, 1900, we find Problem 453 is: "Partager un triangle quelconque en quatre parties équivalentes par deux droites perpendiculaires entre elles!" The problem was solved by L'Hospital before 1704, the year of his death, in a posthumous work, Traité analytique des Sections coniques, Paris, 1707, pp. 400–407. As the result of correspondence inL'Intermédiaire des Math., tomesi–vii, 1894–1900, Questions 3 and 587, Loria wrote the history of the problem: "Osservazioni sopra la storia di un problema pseudo-elementare."Bibl. Math., 1903 (3),iv, 48–51. Leibnitz's name appears in this connection. See note on 1645—C. Huygens. 1688—J. Ozanam. L'usage du compas de proportion expliqué et demontré d'une manière courte et facile, et augmenté d'un Traité de la division des champs. Paris. "Division des champs," pp. 89–138. Ouvrage revu, corrigé et entièrement refondu par J. G. Garnier. Paris, 1794, pp. 165–257. 1694—S. Le Clerc.Traité de Géométrie sur le terrainat end ofGéométrie pratique, ou pratique de la géométrie sur le papier et sur le terrain. Ams- terdam. 1699—J. Ozanam. Cours de mathématique, nouv. éd. tome 3. Paris. Pages23–64. Germantranslation: Anweisung, die geradlinichten Figuren nach einen gegebenen

Verhältniss ohne Rechnung zu theilen. Frankfurt u. Leipzig, 1776. 1704—Guisnée. Application de l'algèbre à la géométrie. Paris. Although the "approbation" signed by Fontenelle is dated "15 Juillet 1704" the work was first published in 1710; second edition "revûe, corrigée et consid- érablement augmentée par l'auteur," Paris, 1733, pp. 42–47; analytic discussion only. 1739— l'abbé Deidier. La science des géométres(sic) ou la théorie et la pra- tique de la géométrie. Paris. "De la géodésie ou division des champs," pp. 279–320; divisions of triangles, rectangles, trapeziums, polygons.

84 EUCLID'S BOOK ON DIVISION OF FIGURES IV 1740—N. Saunderson.Elements of Algebra in ten books, vol. 2. Cambridge. Pages 546–554. 1747—T. Simpson. Elements of Plane Geometry. London. Pages 151–152; new ed., London, 1821, pp. 207–208; taken from Newton (1676). 1748—L. Euler. Introductio in analysin infinitorum. Tomus secundus. Lau- sanne. Chapter 22: "Solutio nonnullorum problematum ad circulum pertinentium." Three of the eight problems which Euler here discusses by the method of trial and error, and tables of circular arcs and logarithmic sines and tangents, are of interest to us. These are: Problem 2, "To find the sector of the circleACB which is divided by the chordAB into two equal parts, so that the triangleACB shall be equal to the segment AEB." Problem 4, "Given the semi-circle AEDB, to draw from the pointA a chord AD which will divide the semi-circle into two equal parts." Problem 5, "From a pointA of the circumference of a circle, to draw two chords AB, AC which shall divide the area of the circle into three equal parts." (Heron, cf. Art. 50.) Gregory (1840) considers these problems at the close (pp. 186–188) of his Appendix. For other editions of Euler's "Introductio," tomus 2, seeVerzeichnis der Schrif- ten Leonhard Eulers.BearbeitetvonG.Eneström. ErsteLieferung, Leipzig,1910. 1752—T. Simpson.Select Exercises for young proficients in the mathematics. London. Problem xlii, pp. 145–6; new ed. by J. H. Hearding. London, 1810, pp. 148–9. 1754—J. le R. D'Alembert. Encyclopédie ou Dictionnaire raisonné des sciences…mis en ordre et publié par M. Diderot…; et quant à la partie mathématique par M. d'Alembert.Paris. Article "Géodésie"; mostly descriptive of methods of Guisnée (1704) and Clerc (1694). 1768—J. A. Euler."AuflösungeinigergeometrischenAufgaben," Abhandlun- gen der Churfürstlich-baierischen Akademie der Wissenschaften, v, 165– 196. Erste Aufgabe, pp. 167–182: "Man soll zeigen, wie eine jede geradlinichte Figur durch Parallellinien in eine gegebene Anzahl gleicher Theile zerschnitten werden kann." Zweite Aufgabe, pp. 182–187: "Eine Zirkel-fläche durch parallellinien in eine gegebene Anzahl gleicher Theile zu zerschneiden." Dritte Aufgabe, pp. 187– 196: "Die Höhe und Grundlinie einer aufrecht-stehenden geschlossenen Para- belfläche ist gegeben, man soll dieselbe durch Parallellinien inn gleiche Theile zerschneiden." Discussion mostly analytic. 1772(?)— J. H. Lamberts deutscher gelehrter Briefwechsel.Herausgegeben von Joh. Bernoulli. Band 2, Berlin, 1782.

APPENDIX 1783–1807 85 Pages 412–13, undated fragment of a letter from Lambert to J. E. Silberschlag. Analytic solution by quadratic equation of the problem: "Ein FeldABCD welches in ABFE Wiesen, in EFCD Ackerfeld ist, soll durch eine gerade LinieKM so getheilt werden, dass so wohl die Wiesen als das Ackerfeld in beliebiger Verhält- niss getheilt werde." [ABCD is a quadrilateral andEF is a straight

line segment joining points on the opposite sidesAD, BC respectively.] In theJournal of the Indian Mathematical Society, 1914,vi, 159, N. P. Pandya proposed as Question 563: “Given two quadrilaterals in the plane of the paper show how to draw a straight line bisecting them both.” A solution by means of common tangents to hyperbolas was offered in 1915,vii, 176. 1783—J. T. Mayer. Gründlicher und ausführlicher Unterricht zur praktis- chen Geometrie,3. Teil. Göttingen. Pages 215–303: “Theilung der Felder durch Rechnung, Theilung der Felder durch blose Zeichnung, Anwendung der Theilungsmethoden auf mancherley, in gemeinen Leben vorkommende Fälle”; dritte Auflage, 1804, pp. 232–337. 1793—J. W. Christiani. Die Lehre von der geometrischen und ökonomis- chen Vertheilung der Felder, nach der Dänischen Schrift des N. Morville bearbeitet von J. W. Christiani.Preface by A. G. Kästner. Göttingen. 1795— Gentleman’s Diary, London. No. 54, 1794, p. 47, Question 691 by J. Rodham: “Within a given triangle to find a point thus, that if lines be drawn from it to cut each side at right angles, the three parts into which the triangle thus becomes divided, shall obtain a given ratio.” Solution by hyperbolas in No. 55, 1795, pp. 37–38. See also Davis’s edition of theGentleman’s Diary, vol. 3, London, 1814, pp. 233–4. 1801—L. Puissant.Recueil de diverses propositions de géométrie résolues ou démontrées par l’analyse algébrique suivant les principes de Monge et de Lacroix. Paris. Pages33–36; Germaned., Berlin, 1806; secondFrenched., Paris, 1809, pp.107–111; third ed., Paris, 1824, pp. 139–142. 1805—M. Hirsch. Sammlung geometrischer Aufgaben, Erster Theil. Berlin. “Theilung der Figuren durch Zeichnung,” pp. 14–25; “Theilung der Figuren durchRechnung,” pp.42–53; Reprint, 1855; EnglisheditiontranslatedbyJ.A.Ross and edited by J. M. F. Wright. London, 1827. 1807—A. Bratt.Problema geometricum triangulum datum a dato puncto in 2 partes aequales secandi. Greifswald. This title is taken fromC. G. Kayser, Bücher-Lexicon, Erster Teil, Leipzig, 1834. 1807—J. P. Carlmark.Triangulus datus a dato puncto in 2 partes aequales secandus. Greifswald.

86 EUCLID’S BOOK ON DIVISION OF FIGURES IV This title and the next two are taken fromE. Wölffing, Math. Bücherschatz, 1903. 1809—J. Kullberg.Problema geometricum triangulum datum e quovis dato puncto in 2 partes aequales secandi. Diss. Lund. 1810—J. Kullberg.Problema geometricum triangulum quodcunque datum in 2 aequales divisum iterum in partes aequales ita secandi, ut rectae secantes angulum constituant rectum. Diss. Upsala. 1811—J. P. Grüson.Geodäsie oder vollständige Anleitung zur geometrischen und ökonomischen Feldertheilung. Halle. 1819—L. Bleibtreu.Theilungslehre oder ausführliche Anleitung, jede Grund- fläche auf die zweckmässigste Art... geometrisch zu theilen. Frankfurt am Main. 1821—J. Leslie. Geometrical Analysis and Geometry of Curve LinesEdin- burgh. Pages 64–66. 1823—A. K. P. von Forstner.Sammlung systematisch geordneter und syn- thetisch aufgelöster geometrischer Aufgaben. Berlin. “TheilungderFlächen, mittelstderProportionundderAehnlichkeit,” pp.310– 371. 1827— Correspondance mathématique et physiquepublié par A. Quetelet, tome iii. Page 180: “On donne dans un plan un angle et un point, et l’on demande de faire passer par le point une droite qui coupe les cotés de l’angle, de manière que l’aire interceptée soit de grandeur donnée.” Solution by Verhulst, pp. 269–270. Answer by Bobillier, tomeiv, pp. 2–3. Generalizing his solution, he gets the result: “tous les plans tangens d’un

hyperboloïde à deux nappes, interceptent sur le cône asymptotique des volumes équivalens." Compare note 117. 1831—P. L. M. Bourdon.Application de l'algèbre à la géométrie comprenant la géométrie analytique à deux et à trois dimensions, troisième édition. Paris. Pages 46–54; 5e éd., Paris, 1854, pp. 33–41; 8e éd. rev. par Darboux, Paris, 1875, pp. 30–38. Analytic discussion only. 1831—H. v. Holleben, und P. Gerwien. Geometrische Analysis. Berlin, 2 Bde, 1831–1832. "Theilungen," i, 184–191;ii, 144–151. 1837—G. Ritt. Problèmes d'applications de l'algèbre à la géométrie avec les solutions développées, 2e partie. Paris. Pages 108–109.

APPENDIX 1840–1857 87 1840—O. Gregory.Hints theoretical, elucidatory and practical, for the use of Teachers of elementary Mathematics and of self-taught students; with es- pecial reference to the first volume of Hutton's course and Simson's Euclid, as Text-Books. Also a selection of miscellaneous tables, and an Appendix on the geometrical division of plane surfaces. London. "Appendix: Problems relative to the division of Fields and other surfaces," pp. 158–188; partly taken from Hirsch (1805). See also Euler (1748). 1844—Dreser. Die Teilung der Figuren. Darmstadt. This title is taken fromE. Wölffing, Math. Bücherschatz, 1903. 1847—R. Potts. An appendix to the larger edition of Euclid's Elements of Geometry; containing...Hints for the solution of the Problems...Cam- bridge and London. Ex. 91, pp. 72–73. 1852—H. Ch. de La Frémoire.Théorèmes et Problèmes de Géométrie élémentaire, seconde éd. revue et corrigée par E. Catalan. Paris. Pages 107–108; 6e éd. par Catalan, Paris, 1879, pp. 190–191. 1852—F. Rummer.Die Verwandlung und Theilung der Flächen in einer Reihe von Constructions- u. Berechnungs-Aufgaben. Mit 3 Steintafeln. Heidel- berg. 6 + 90 pp. 1855—P. Kelland. "On Superposition." Transactions of the Royal Society of Edinburgh, 1885,xxi, 271–273 + 1 pl. This paper deals, for the most part, with solutions of the following problem proposed to Professor Kelland by Sir John Robison: "From a given square one quarter is cut off, to divide the remaining gnomon into four such parts that they shall be capable of forming a square." In theTransactions, 1891,xxxvi, 91–95, + 2 pls., Robert Brodie has a paper entitled "Professor Kelland's Problem on Superposition." 1857—E. Catalan. Manuel des Candidats à l'école polytechnique. Paris, Tome I. Pages 233–4: "To divide a circle into two equal parts by means of an arc with its centre, A, on the circumference of the given circles." This is stated by A. Rebière (Mathématique et Mathématiciens, 2e éd., Paris, 1893, p. 519) under the form: "Quelle doit être la longueur de la longe d'un cheval pour qu'en la fixant au contour d'un pré circulaire l'animal ne puisse tondre que la moitié du pré?" The solution of this problem leads to a transcendental equation $\sin x - x\cos x = \pi 2$,

88 EUCLID'S BOOK ON DIVISION OF FIGURES IV where x is the angle under which the points of section of the circumferences are seen fromA. Catalan findsx = 109◦11′18″, correct to within a second of arc. Cf. L'Intermédiaire des Mathématiciens, 1914, Question 4327,xxi, 5, 69, 90, 115, 180. 1863—J. McDowell. Exercises on Euclid and in Modern Geometry. Cam- bridge. No. 157, pp. 145–6; 3rd ed. 1881, p. 118. 1864— Educational Times Reprint, Vols. 1, 40, 44, 66, 68, 69; new series, Vol. 1; 1864–1910. The problems here solved are Euclid's 19, 20, 26, 27: No. 1457 (i, 49, old edi- tion, 1864) proposed by R. Palmer; solution by Rutherford who states that it was also published in Thomas Bradley'sElements of Geometrical

Drawing, 1861— Nos. 7336 and 7369 (xl, 39, 1884) proposed by W. H. Blythe and A. H. Cur- tis; solutions by G. Heppel and Matz—No. 8272 (xliv, 92, 1886) proposed by E. Perrin; solution by D. Biddle—No. 12973 (lxvi, 29, 1897) proposed by Rad- hakrishnan; solution by I. Arnold—No. 13460 (lxviii, 35, 1898) proposed by I. Arnold; solution by W. S. Cooney, etc.—No. 13647 (lxix, 42, 1898) proposed by I. Arnold; solution by W. C. Stanham—No. 16747 (new seriesxviii, 46, 1910) proposed by I. Arnold; solution by proposer, by Euclid'sElements Bk i. 1864—H. Hölscher.Anleitung zur Berechnung und Teilung der Polygone bei rechtwinkligen Koordinaten. Berlin and Charlottenburg. This work and the two following are representative of those which treat of Divisions of Figures by computation, rather than by graphical methods: (1) F. G. Gauss, Die Teilung der Grundstücke, insbesondere unter Zugrundelegung Koordinaten, 2 Auflage, Berlin, 1890; (2)L. Zimmerman, Tafeln für die Teilung der Dreiecke, Vierecke, und Polygone, Zweite vermehrte und verbesserte Auflage, Liebenwerda, 1896; 118+64 pp. 1870—F. Lindman. "Problema geometricum." Archiv der Math. u. Phys. (Grunert), Bd 51, 1870, pp. 247–252. 1879—P. M. H. Laurent.Traité d'algèbre à l'usage des candidats....Troi- sième édition. Paris. Tome I, p. 191: "To divide a triangle into two equal parts by the shortest possible line." Solutions inL'Intermédiaire des Mathématiciens, 1902, IX, 194–5. See also F. G. M.,Exercices de Géométrie, Cinquième édition. Tours et Paris, 1912, p. 802. 1892—H. S. Halland F. H. Stevens. Key to the Exercises and Examples contained in a Text-Book of Euclid's Elements. London. Ex. 7, 8, 10, 11, pp. 163–164.

APPENDIX 1894–1911 89 1894—G. E. Crawford."Geometrical Problem." Proc. Edinb. Math. Soc., Vol. 13, 1895, p. 36. Paper read Dec. 14, 1894. 1899—W. J. Dilworth.A New Sequel to Euclid. London. Ex. xxxv, p. 190. 1901—A. Larmor.Geometrical Exercises from Nixon's 'Euclid Revised.'Ox- ford. Ex. 15, p. 122. 1902—C.Smith. Solution of the Problems and Theorems in Smith and Bryant's Elements of Geometry. London. Ex. 121, pp. 177–178; T. Simpson's solution and another. 1910—H. Flükiger. Die Flächenteilung des Dreiecks mit Hilfe der Hyperbel. Diss. Bern. 50 pp. + 3 plates. 1910—R. Zdenek."Halbierung der Dreiecksfläche." Wien,Zeitschrift für das Realwesen, Jahrgangxxxv, Heft 10, 8 pp. Discussion by projective geometry leading to hyperbolic arcs. 1911—D. Biddle. Problem 17197, Educational Times, London, November, lxiv, 475. "Divide a square into five right-angled triangles, the areas of which shall be in arithmetic progression." Solutions in the Educational Times Reprint, new series, xxvi, iii, 1914.

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